

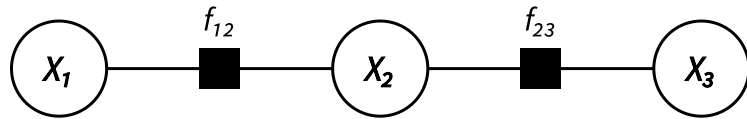
Sampling replaces an impossible sum

Maximum entropy, moment matching and Markov chain Monte Carlo

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Week 3 ended with a model we can write down but cannot yet use



$$f_{12}(x_1, x_2)$$

	$x_2 = 0 \ 1$	
$x_1 = 0$	4	1
$x_1 = 1$	8	5

$$f_{23}(x_2, x_3)$$

	$x_3 = 0 \ 1$	
$x_2 = 0$	3	1
$x_2 = 1$	1	4

$$g(x) = f_{12}(x_1, x_2) \cdot f_{23}(x_2, x_3)$$

$$p(x) = g(x) / Z, Z = 78$$

- **Energy.** A positive table is $e^{-\text{energy}}$: $E(x) = -\ln g(x)$.
- **Boltzmann distribution.** $p_T(x) = e^{-E(x)/T} / Z_T$, temperature T .
- **Boltzmann machine.** Spins $s_i = \pm 1$, couplings J_{ij} , fields h_i .
- **Open question 1.** Every energy was *given*. Why this exponential form, and how are h , J fitted to data?
- **Open question 2.** Every answer was an exact sum over all 2^n states.

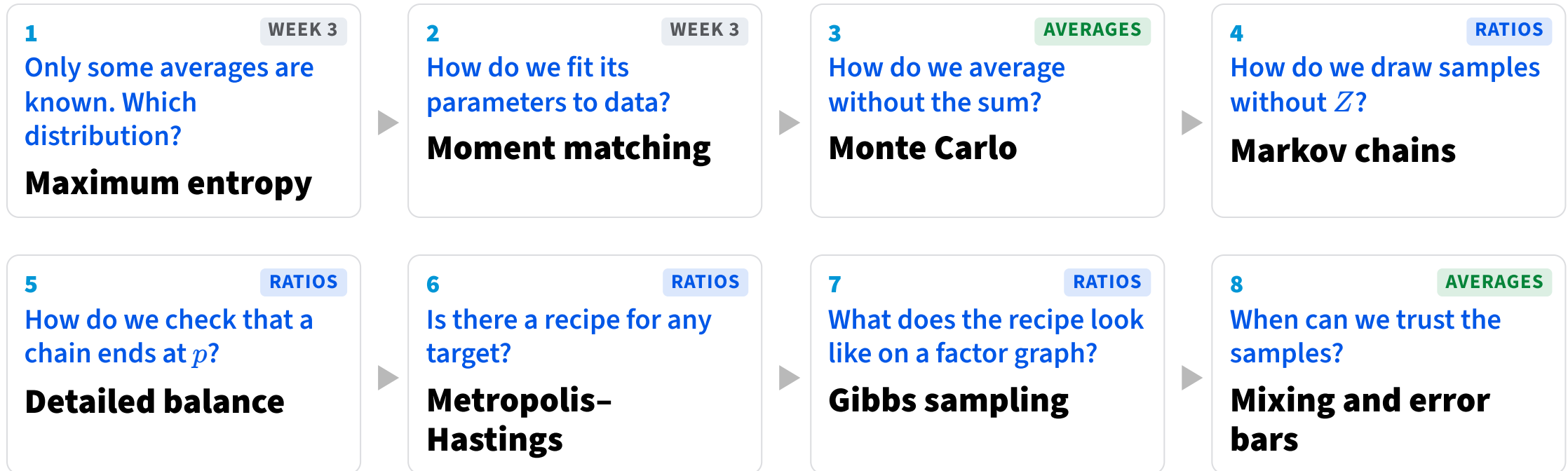
Week 3's toolbox: one quantity answers each question

Question	Quantity	How to read it
How uncertain is a variable X ?	entropy $H(X) = -\sum_x p(x) \ln p(x)$	0 when X is certain; largest, $\ln K$, when all K values are equally likely
How much does observing Y tell us about X ?	mutual information $I(X; Y) = H(X) - H(X Y)$	the drop in uncertainty about X ; 0 when the two are independent
A model predicts q , the data come from p . How well does q predict?	log loss (cross-entropy) $H(p, q) = -\sum_x p(x) \ln q(x)$	smaller is better; the lowest value, $H(p)$, only for $q = p$
How much does a wrong q cost?	KL divergence $D_{\text{KL}}(p \parallel q) = H(p, q) - H(p)$	the extra log loss; never negative, 0 only for $q = p$
Which configuration is most likely?	energy $E(x) = -\ln g(x)$	lower energy is more likely: $p(x) \propto e^{-E(x)}$
What does the temperature T do?	$p_T(x) \propto e^{-E(x)/T}$	small T : probability piles onto the lowest energies; large T : it spreads out

Today: **which model to choose**, and **how to average without exact sums**.

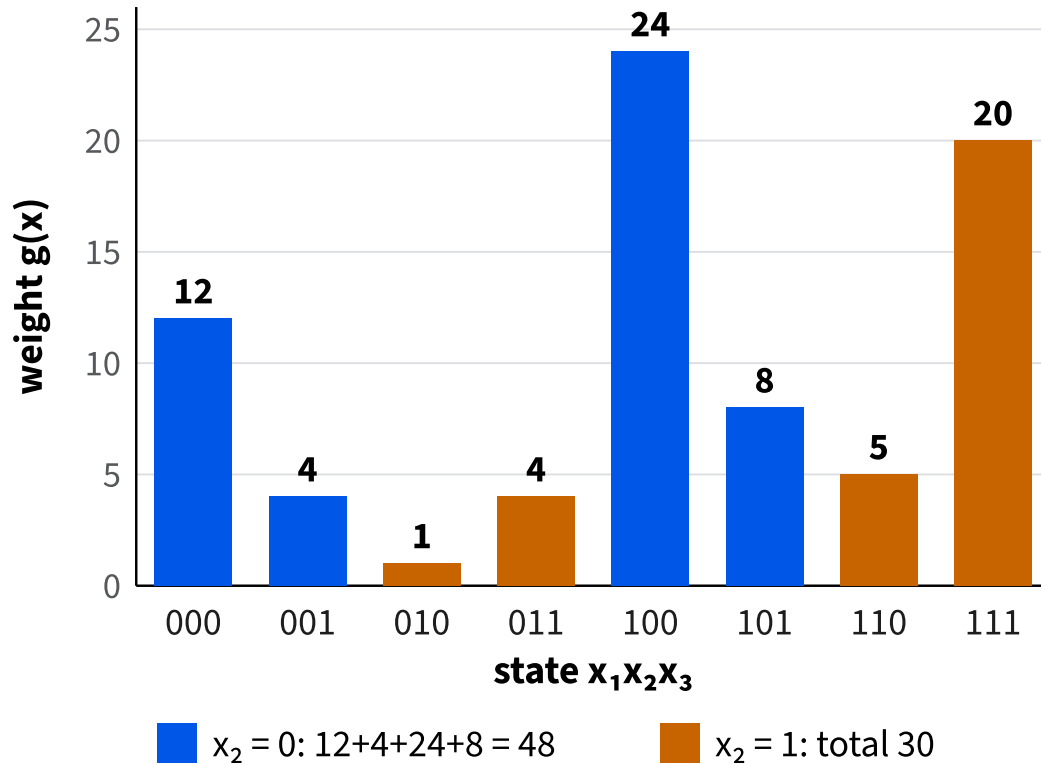
One story in eight questions: each answer raises the next question

One number blocks every use of the model: the normalizer Z . There are two ways around it: **averages** and **ratios**.



Next week: **learning** with sampled averages, and **expectation-maximization (EM)** when some variables are hidden.

One model carries the whole lecture: the Week 2 chain



- Chain $X_1 - f_{12} - X_2 - f_{23} - X_3$, three binary variables.
- Weight $g(x) = f_{12}(x_1, x_2) f_{23}(x_2, x_3)$; normalizer $Z = 78$.
- **Week 2's answer:** $P(X_2 = 0) = 48/78 = 0.6154$, computed exactly by messages.
- **Eight states:** every method today is checked against these exact numbers.
- **Later:** a 32×32 grid of spins has $2^{1024} \approx 10^{308}$ states. No exact answer exists, yet the samplers still work.

Every concept follows the same five steps

Idea. A small example on the chain, with a picture or an animation. You see the method work before it has a name.

Definition and theorem. The precise statement, with a proof you can follow line by line.

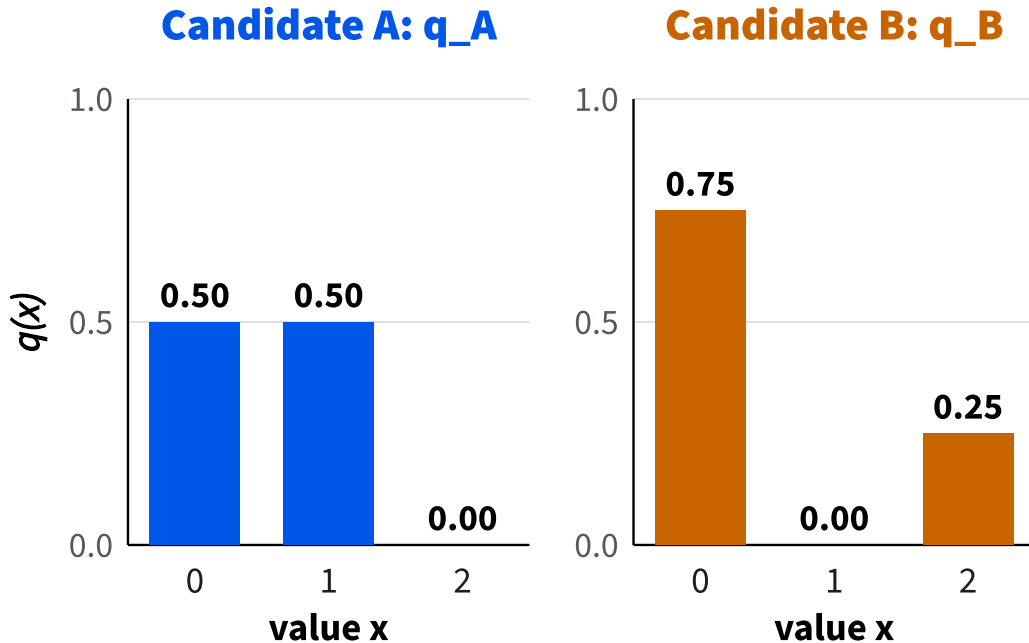
Check. The statement applied to the chain, with exact numbers.

In today's machine learning. Where the same idea runs in current systems, with the paper.

Connection. What the concept cannot do yet: the next question, which the next concept answers.

The rail at the bottom of every slide shows which of the eight concepts we are in.

Only an average is known, and many distributions fit it



- $X \in \{0, 1, 2\}$; we know only $\mathbb{E}[X] = 0.5$.
- Candidate A: $q_A = (0.5, 0.5, 0)$. Candidate B: $q_B = (0.75, 0, 0.25)$. Both have mean 0.5.
- q_A says $X = 2$ never happens; q_B says $X = 1$ never happens. **The data support neither claim.**

Which distribution should we believe?

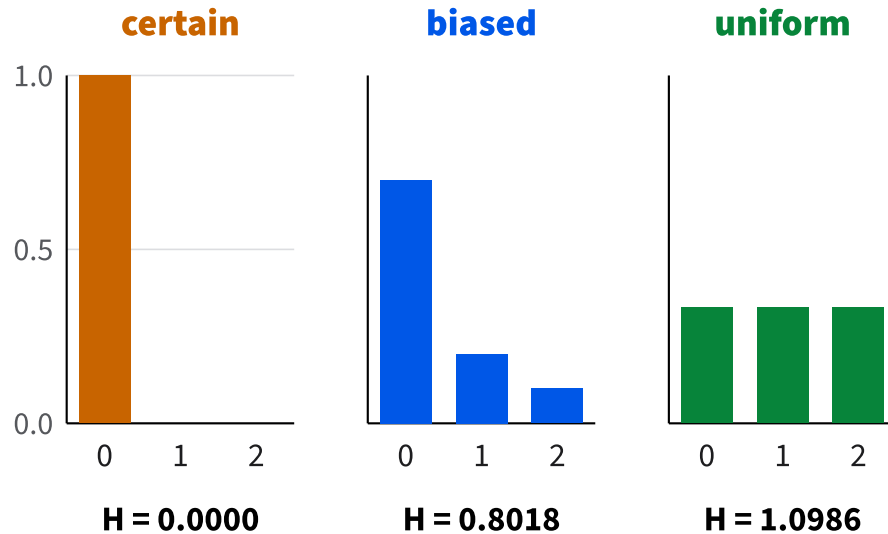
☐ A

☐ B

☒ Something in between

Answer. Something in between: the most spread-out distribution with mean 0.5. Week 3 measured spread by **entropy**.

With no information at all, the uniform distribution has the largest entropy



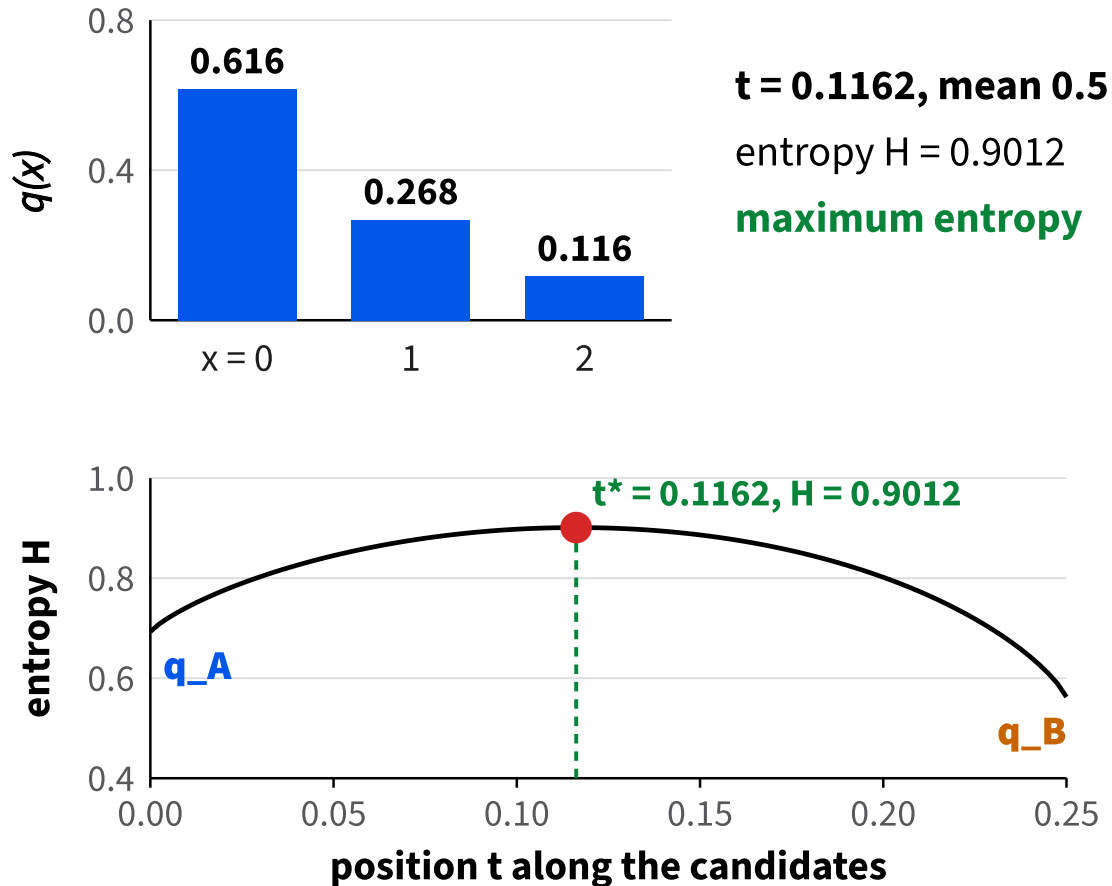
maximum $\ln 3 = 1.0986$, reached only by the uniform distribution

- K values, only the constraint $\sum_x q(x) = 1$.
- Compare q with the uniform $u(x) = 1/K$ by Week 3's KL divergence:

$$D_{\text{KL}}(q \parallel u) = \sum_x q(x) \ln \frac{q(x)}{1/K} \\ = \ln K - H(q) \geq 0.$$

- So $H(q) \leq \ln K$, with equality only when $q = u$.
- **Pattern:** an entropy gap equals a KL divergence. It proves every result in this part.

Slide through every candidate with mean 0.5: the entropy peaks at one of them



- Every candidate: $q_t = (0.5 + t, 0.5 - 2t, t)$, $0 \leq t \leq \frac{1}{4}$. $t = 0$ gives q_A , $t = \frac{1}{4}$ gives q_B .
- Each has mean $(0.5 - 2t) + 2t = 0.5$.
- Press **Play**: the entropy rises, then falls.
- **Peak**: $t^* = 0.1162$, $p^* = (0.6162, 0.2676, 0.1162)$, $H = 0.9012$.
- Every value stays possible; larger values are rarer.

Where the peak is: set the slope of the entropy to zero

$$H(t) = -(0.5 + t) \ln(0.5 + t) - (0.5 - 2t) \ln(0.5 - 2t) - t \ln t.$$

$$H'(t) = -\ln(0.5 + t) - 1 + 2\ln(0.5 - 2t) + 2 - \ln t - 1 = \ln \frac{(0.5 - 2t)^2}{(0.5 + t)t}.$$

Each term $-c \ln c$ has derivative $-(\ln c + 1) c'$; the three constants $-1 + 2 - 1$ cancel.

$$H'(t) = 0 \iff (0.5 - 2t)^2 = (0.5 + t)t \iff 3t^2 - 2.5t + 0.25 = 0.$$

$$t = \frac{2.5 \pm \sqrt{2.5^2 - 3}}{6} = \frac{5 \pm \sqrt{13}}{12} : \quad t^* = \frac{5 - \sqrt{13}}{12} = 0.1162 \quad (\text{the other root } 0.7171 > \tfrac{1}{4}).$$

$p^* = (0.5 + t^*, 0.5 - 2t^*, t^*) = (0.6162, 0.2676, 0.1162)$. H is concave, so this is the maximum.

The winner is a Boltzmann distribution

The peak condition $(0.5 - 2t)^2 = (0.5 + t)t$ says two ratios are equal:

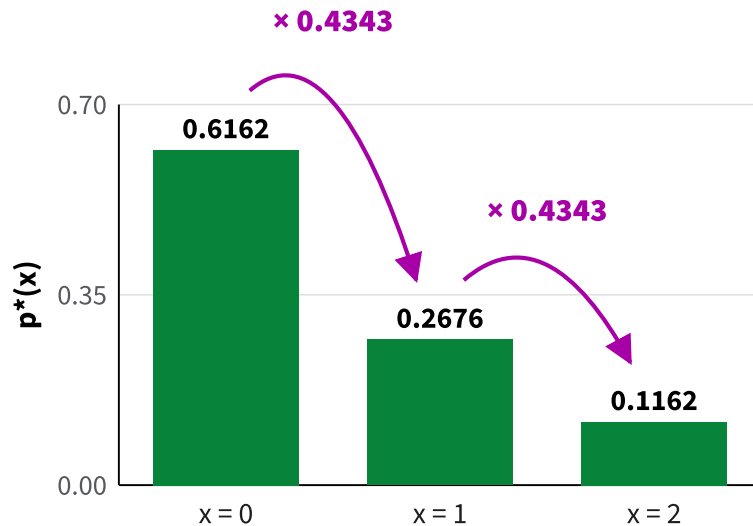
$$\frac{p^*(1)}{p^*(0)} = \frac{0.5 - 2t^*}{0.5 + t^*} = 0.4343,$$

$$\frac{p^*(2)}{p^*(1)} = \frac{t^*}{0.5 - 2t^*} = 0.4343.$$

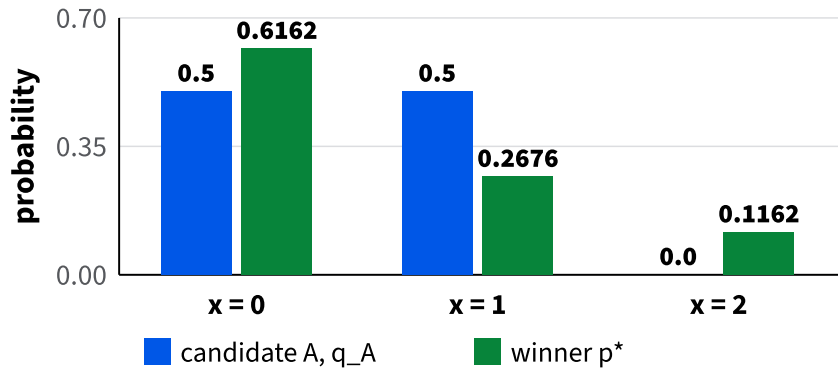
Each step up in x multiplies the probability by $r = 0.4343$. Writing $r = e^{-1/T}$ gives $T = -1/\ln r = 1.20$:

$$p^*(x) = \frac{e^{-x/T}}{Z}, \quad T = 1.20.$$

Boltzmann, with energy $E(x) = x$.



Candidate A's entropy gap is a KL divergence



- Candidate A from the first slide of this part: $q_A = (0.5, 0.5, 0)$.
- The winner: $p^* = (0.6162, 0.2676, 0.1162)$.
- Both have mean 0.5. Since $q_A(2) = 0$, the $x = 2$ terms of q_A vanish below.

$$H(q_A) = -(0.5 \ln 0.5 + 0.5 \ln 0.5) = \ln 2 = 0.6931,$$

$$H(p^*) = -(0.6162 \ln 0.6162 + 0.2676 \ln 0.2676 + 0.1162 \ln 0.1162) = 0.9012,$$

$$H(p^*) - H(q_A) = 0.2081.$$

$$D_{\text{KL}}(q_A \parallel p^*) = 0.5 \ln \frac{0.5}{0.6162} + 0.5 \ln \frac{0.5}{0.2676} = -0.1045 + 0.3126 = 0.2081.$$

The same number. The next slides prove this holds for every constraint.

Which temperature gives a required average energy?

Problem. Given an energy E and a target U : maximize $H(q)$ over all q with $\mathbb{E}_q[E] = U$. Theorem 1 below shows the answer is $p_T \propto e^{-E/T}$, with the T whose mean energy equals U .

$$U(T) = \mathbb{E}_{p_T}[E] = \sum_x E(x) p_T(x) = \frac{\sum_x E(x) e^{-E(x)/T}}{\sum_x e^{-E(x)/T}}.$$

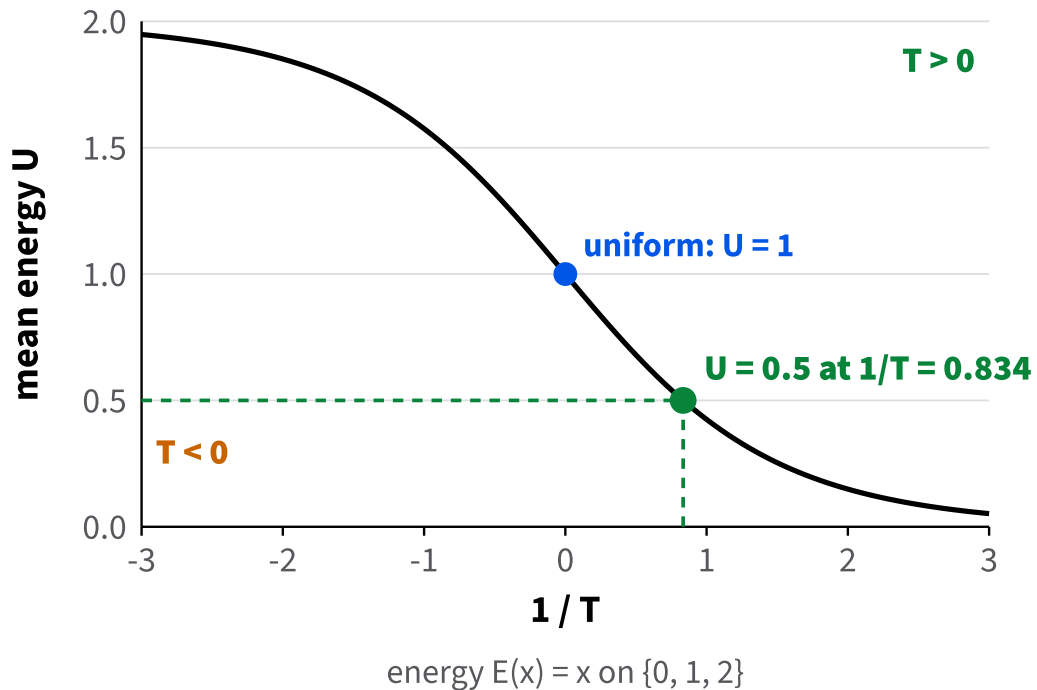
Our example: $E(x) = x$ on $\{0, 1, 2\}$. Write $r = e^{-1/T}$, so that $e^{-x/T} = r^x$:

$$U(T) = \frac{0 \cdot 1 + 1 \cdot r + 2 \cdot r^2}{1 + r + r^2} = \frac{r + 2r^2}{1 + r + r^2}.$$

$$U(T) = 0.5 \iff r + 2r^2 = 0.5(1 + r + r^2) \iff 3r^2 + r - 1 = 0 \iff r = \frac{\sqrt{13}-1}{6} = 0.4343.$$

So $T = -1/\ln r = 1.20$, and r is the ratio 0.4343 found in p^* .

The mean energy falls as $1/T$ grows



- The curve is $U(T) = \frac{r + 2r^2}{1 + r + r^2}$, $r = e^{-1/T}$, plotted against $1/T$.
- $1/T \rightarrow +\infty$ ($T \rightarrow 0^+$): all probability on $x = 0$, so $U \rightarrow 0$.
- $1/T = 0$ ($T = \infty$): uniform, $U = 1$.
- $1/T < 0$ ($T < 0$): weights grow with energy, $U \rightarrow 2$.
- **Only targets below the uniform average need a positive temperature.** Ours, $U = 0.5$, gives $1/T = 0.834$.

Three facts from Week 3 are all the proof needs

Fact 1 (surprise). The logarithm of $p_T = e^{-E/T} / Z_T$:

$$-\ln p_T(x) = \frac{E(x)}{T} + \ln Z_T.$$

Fact 2 (entropy). Average Fact 1 under p_T ; write $U = \mathbb{E}_{p_T}[E]$:

$$H(p_T) = \frac{U}{T} + \ln Z_T.$$

Fact 3 (cross-entropy). For distributions q and p , the **cross-entropy** is the average surprise $-\ln p(x)$ when x is drawn from q (Week 3's log loss). Add and subtract $\sum_x q(x) \ln q(x)$:

$$H(q, p) = - \sum_x q(x) \ln p(x) = - \underbrace{\sum_x q(x) \ln q(x)}_{H(q)} + \underbrace{\sum_x q(x) \ln \frac{q(x)}{p(x)}}_{D_{\text{KL}}(q \parallel p)}.$$

Gibbs' inequality (Week 3): $D_{\text{KL}}(q \parallel p) \geq 0$, with equality only when $q = p$.

At a fixed average energy, the Boltzmann distribution has the largest entropy

Theorem 1. Fix an energy E and a temperature T . Let p_T be the Boltzmann distribution

$$p_T(x) = \frac{e^{-E(x)/T}}{Z_T}, \quad Z_T = \sum_x e^{-E(x)/T},$$

and $U = \mathbb{E}_{p_T}[E]$ its average energy. Every q with $\mathbb{E}_q[E] = U$ has $H(q) \leq H(p_T)$, with equality only for $q = p_T$.

Proof. Average Fact 1 under q , not under p_T . Since q has the same mean energy U :

$$H(q, p_T) = - \sum_x q(x) \ln p_T(x) = \sum_x q(x) \left(\frac{E(x)}{T} + \ln Z_T \right) = \frac{U}{T} + \ln Z_T = H(p_T) \quad (\text{Fact 2}).$$

$$H(p_T) - H(q) = H(q, p_T) - H(q) = D_{\text{KL}}(q \parallel p_T) \geq 0 \quad (\text{Fact 3}). \quad \blacksquare$$

Check: the gap 0.2081 of candidate A is this D_{KL} , with $q = q_A$ and $p_T = p^*$.

With several constrained averages, the answer is an exponential family

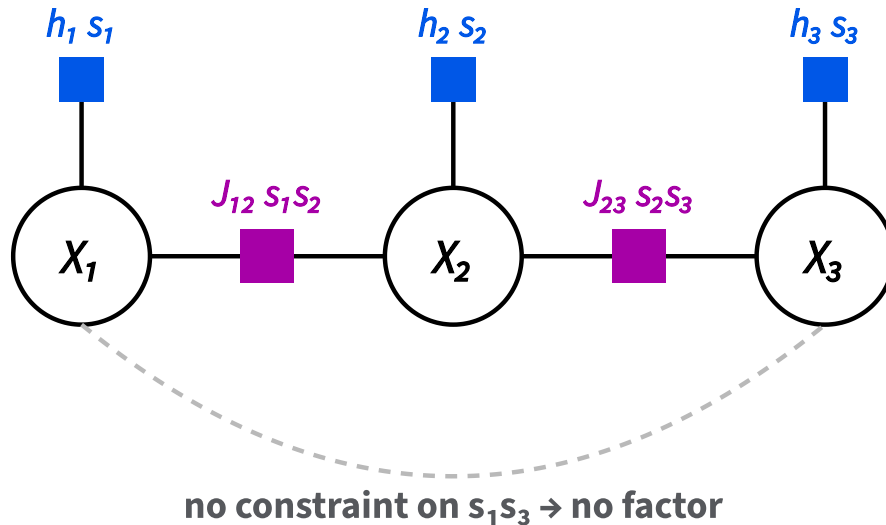
Definition (maximum-entropy problem). Features $\phi_1, \dots, \phi_K$ and measured averages $m_1, \dots, m_K$:

$$\max_q H(q) \quad \text{subject to} \quad \mathbb{E}_q[\phi_k] = m_k \quad (k = 1, \dots, K), \quad \sum_x q(x) = 1.$$

Theorem 2. If some $p_\theta(x) = \exp(\sum_k \theta_k \phi_k(x)) / Z(\theta)$ has the averages m_k , it is the unique maximizer.

Proof. Theorem 1 with $-E/T$ replaced by $\sum_k \theta_k \phi_k$: every feasible q has $H(q, p_\theta) = -\sum_k \theta_k m_k + \ln Z(\theta) = H(p_\theta)$, so $H(p_\theta) - H(q) = D_{\text{KL}}(q \parallel p_\theta) \geq 0$. ■

On our chain, each constrained average becomes a factor



= the chain $X_1 - X_2 - X_3$

- Spins $s_i = 2x_i - 1 \in \{-1, +1\}$. Suppose we know only five averages of the chain.
- **Unary**: $\mathbb{E}[s_1] = \frac{6}{13}$, $\mathbb{E}[s_2] = -\frac{3}{13}$, $\mathbb{E}[s_3] = -\frac{1}{13}$. Each gives a factor $e^{h_i s_i}$ on one variable.
- **Pairs**: $\mathbb{E}[s_1 s_2] = \frac{2}{39}$, $\mathbb{E}[s_2 s_3] = \frac{7}{13}$. Each gives a factor $e^{J_{ij} s_i s_j}$ on two variables.
- No average of $s_1 s_3$ was imposed, so X_1 and X_3 share **no** factor.
- **The constraints draw the factor graph.**

The five averages of our chain give back exactly our chain

$$p(s) = \frac{1}{Z} \exp(h_1 s_1 + h_2 s_2 + h_3 s_3 + J_{12} s_1 s_2 + J_{23} s_2 s_3)$$

Solving the five average equations gives the Ising form of Week 3:

$$h = (0.5756, -0.3922, 0.0719), \quad J_{12} = \frac{1}{4} \ln 2.5 = 0.2291, \quad J_{23} = \frac{1}{4} \ln 12 = 0.6212.$$

Check one ratio. From 000 to 100 only s_1 changes, from -1 to $+1$:

$$\frac{p(100)}{p(000)} = e^{2h_1 - 2J_{12}} = e^{2(0.5756 - 0.2291)} = e^{0.6931} = 2 = \frac{24}{12}. \checkmark$$

Without constraints, $\ln Z$ is the best entropy–energy trade-off

Theorem 3 (variational form). As before, $p_T(x) = e^{-E(x)/T} / Z_T$ with $Z_T = \sum_x e^{-E(x)/T}$. For every distribution q ,

$$H(q) - \frac{\mathbb{E}_q[E]}{T} \leq \ln Z_T, \quad \text{with equality only for } q = p_T.$$

So $\ln Z_T$ is a maximum over all distributions:

$$\ln Z_T = \max_q \left\{ H(q) - \frac{\mathbb{E}_q[E]}{T} \right\}.$$

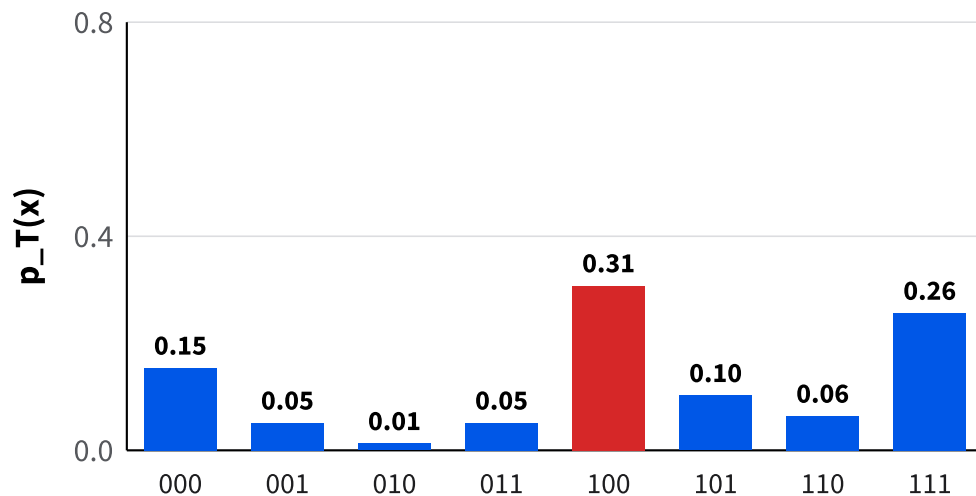
Proof. Expand the KL divergence with Fact 1:

$$D_{\text{KL}}(q \parallel p_T) = \sum_x q(x) \ln q(x) - \sum_x q(x) \ln p_T(x) = -H(q) + \frac{\mathbb{E}_q[E]}{T} + \ln Z_T \geq 0. \quad \blacksquare$$

We return to this formula in Week 6: when Z cannot be summed, maximizing over a simpler family of q gives a bound.

Temperature trades entropy against energy

Global function of the chain: $g(x) = f_{12}(x_1, x_2) f_{23}(x_2, x_3)$, with values 12, 4, 1, 4, 24, 8, 5, 20 for $x = 000, \dots, 111$.



T = 1.00: entropy H = 1.7698

mean energy = -2.5869, $\ln Z_T = 4.3567$

$$p_T(x) = \frac{g(x)^{1/T}}{\sum_{x'} g(x')^{1/T}} = \frac{e^{-E(x)/T}}{Z_T}$$

where $E = -\ln g$. At $T = 1$ this is $g(x)/78$, the chain itself; at $T = \frac{1}{2}$ it is $g(x)^2/1242$, so $p_T(100) = \frac{576}{1242} = 0.46$.

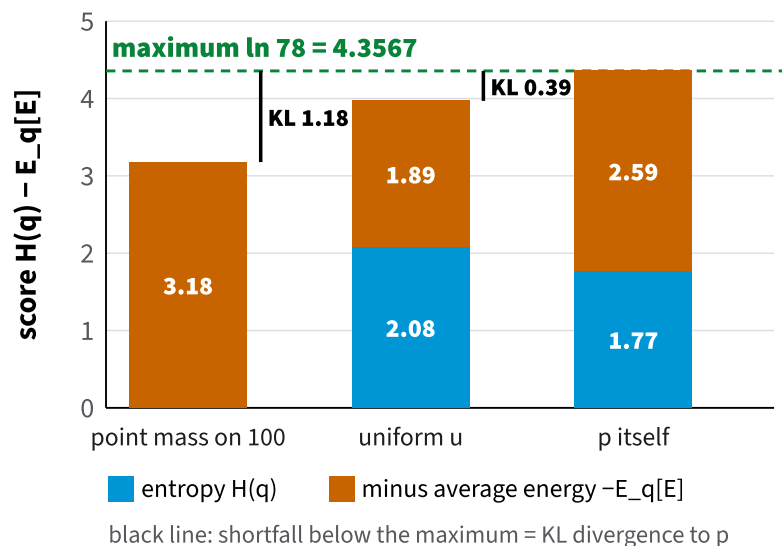
- **Low T** : energy wins; probability piles on 100.
 $T = 0.25$: $p_T(100) = 0.64$.
- **High T** : entropy wins; probability spreads.
 $T = 4$: $p_T(100) = 0.17$.
- $T \rightarrow \infty$: uniform, $H = \ln 8$. Theorem 3: p_T is the best compromise at each T .

Scoring three trial distributions on the chain

Global function $g(x) = f_{12}(x_1, x_2) f_{23}(x_2, x_3)$, energy $E(x) = -\ln g(x)$, $T = 1$. Score of q : $H(q) - \mathbb{E}_q[E]$.

Theorem 3: the best score is $\ln Z = \ln 78 = 4.3567$.

$x = x_1 x_2 x_3$	000	001	010	011	100	101	110	111
$g(x)$	12	4	1	4	24	8	5	20
$E(x) = -\ln g(x)$	-2.48	-1.39	0	-1.39	-3.18	-2.08	-1.61	-3.00

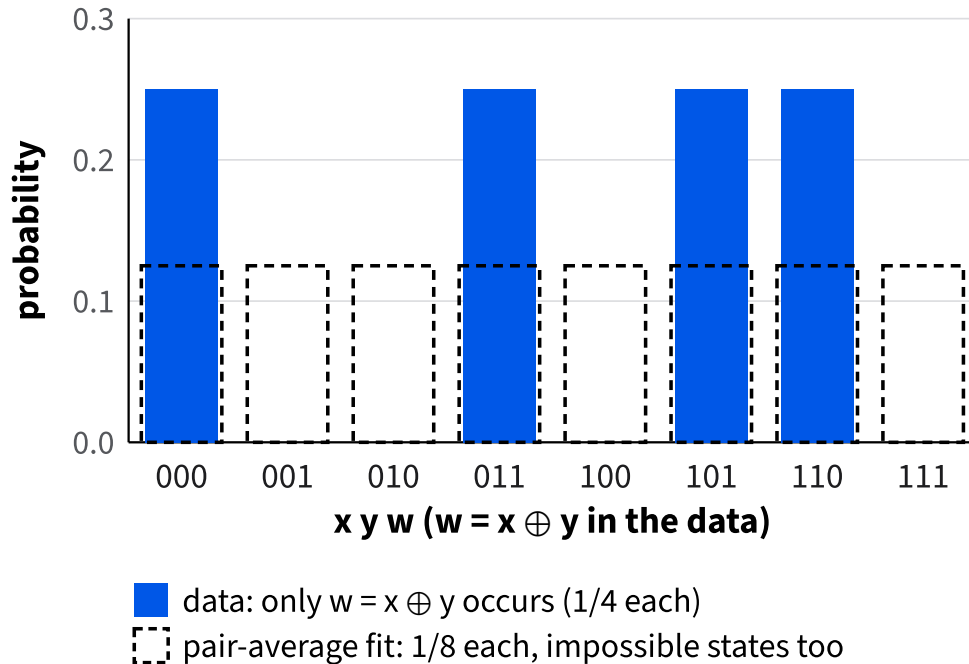


Point mass on 100. $H = 0$, $\mathbb{E}[E] = E(100) = -\ln 24$: score $\ln 24 = 3.1781$; short by $\ln \frac{78}{24} = 1.1787$.

Uniform u . $H = \ln 8 = 2.0794$, $\mathbb{E}_u[E] = \frac{1}{8} \sum_x E(x) = -1.8900$: score 3.9695; short by $0.3872 = D_{\text{KL}}(u \parallel p)$.

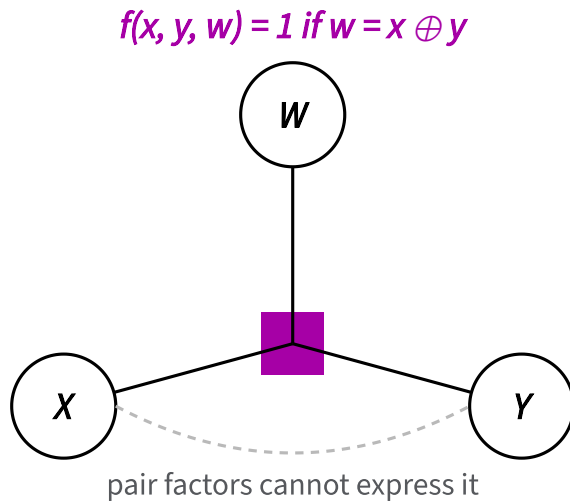
$p = g/78$ itself. $H(p) = 1.7698$, $\mathbb{E}_p[E] = -2.5869$: score $4.3567 = \ln 78$ (Fact 2: $H(p) - \mathbb{E}_p[E] = \ln Z$). **Shortfall 0.**

Pair averages cannot see a rule that involves three variables



- Week 3's XOR: X, Y fair coins, $W = X \oplus Y$. Only four states occur, $xyw = 000, 011, 101, 110$, each $\frac{1}{4}$.
- **One variable at a time:** each of X, Y, W is 1 in two of the four states, so each is 1 with probability $\frac{1}{2}$.
- **Two at a time:** each pair, e.g. (X, W) , takes the values 00, 01, 11, 10 once each, so each combination has probability $\frac{1}{4}$.
- Three **independent** fair coins give exactly the same one- and two-variable statistics. Only all three together reveal $W = X \oplus Y$.
- So maximum entropy with these constraints returns **three independent coins, uniform on all 8 states.**
The four impossible states get $\frac{1}{8}$ each.

Which local function restores the missing rule?



Pair factors cannot express $W = X \oplus Y$. What can?

A stronger pair factor

One factor on all three

More data

Answer. One square joined to all three circles:

$f(x, y, w) = 1$ if $w = x \oplus y$, else 0. Its feature $s_x s_y s_w$ has data average -1 .

- With this factor the model keeps exactly the four possible states: $H = \ln 4$.
- Those states, 000, 011, 101, 110, each contain an **even number of ones**. So the factor is a **parity check**: 1 for an even count, 0 for an odd count.
- The same check works for more bits. On four bits it keeps the 8 of 16 states with an even count, such as 0000, 0011, 1111, and removes the other 8.

Maximum entropy is a modeling principle, not a guarantee about the world

Fitting p_θ to measured averages makes three different claims:

Claim 1. Its averages equal the measured ones. **Provable:** this is how θ is chosen.

Claim 2. Among all distributions with those averages, it has the largest entropy. **Provable:** Theorem 2.

Claim 3. It matches the real system. **Not provable.** XOR satisfies Claims 1 and 2 and still puts $\frac{1}{2}$ on impossible states.

So test a fitted model on a property you did not use to fit it. An omitted relationship stays omitted, however exact the arithmetic.

The softmax classifier is the maximum-entropy model

$$p_{\theta}(y \mid x) = \frac{\exp \left(\sum_k \theta_k \phi_k(x, y) \right)}{\sum_{y'} \exp \left(\sum_k \theta_k \phi_k(x, y') \right)}$$

- Features $\phi_k(x, y)$ of an input x and a label y .
- Theorem 2 applied to each input: the maximum-entropy model with those averages **is a softmax**.

- Text classifiers were built this way in the 1990s (Berger, Della Pietra and Della Pietra, 1996).
- A language model's output layer is the same form: logits z_y , probabilities $e^{z_y} / \sum_{y'} e^{z_{y'}}$ over the vocabulary.
- The denominator is a **small** sum here, one term per token. It becomes impossible when y is a whole response or an image.

A. L. Berger, S. A. Della Pietra, V. J. Della Pietra. A maximum entropy approach to natural language processing. *Computational Linguistics* 22(1), 39–71, 1996.

Alignment with human feedback solves Theorem 3's maximization over responses

Our chain

A chat model

state x	response y
energy $E(x)$	minus reward $-r(y)$
temperature T	KL weight β
uniform base	reference model π_{ref}
$p_T \propto e^{-E/T}$	$\pi^* \propto \pi_{\text{ref}} e^{r/\beta}$

R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, C. Finn.
Direct preference optimization: your language model is secretly a
reward model. *NeurIPS* 2023.

- **Reinforcement learning from human feedback (RLHF):** $\max_{\pi} \mathbb{E}_{\pi}[r(y)] - \beta D_{\text{KL}}(\pi \parallel \pi_{\text{ref}})$.
- Divided by β , this is Theorem 3 with a reference distribution. The maximizer is **Boltzmann over responses:** $\pi^*(y) = \pi_{\text{ref}}(y) e^{r(y)/\beta} / Z$.
- $Z = \sum_y \pi_{\text{ref}}(y) e^{r(y)/\beta}$ sums over **every possible response**: out of reach.
- So how can a model be trained toward π^* ?
The next two slides.

Compare two responses, and Z cancels

$$\pi^*(y) = \frac{\pi_{\text{ref}}(y) e^{r(y)/\beta}}{Z} \quad \text{for every response } y, \text{ with the same } Z.$$

Divide the formula for one response y_1 by the formula for another response y_2 :

$$\frac{\pi^*(y_1)}{\pi^*(y_2)} = \frac{\pi_{\text{ref}}(y_1) e^{r(y_1)/\beta} / \cancel{Z}}{\pi_{\text{ref}}(y_2) e^{r(y_2)/\beta} / \cancel{Z}} = \frac{\pi_{\text{ref}}(y_1)}{\pi_{\text{ref}}(y_2)} e^{(r(y_1) - r(y_2))/\beta}.$$

Toy check. Three responses, $\pi_{\text{ref}} = (0.5, 0.3, 0.2)$, rewards $r = (1, 2, 0)$, $\beta = 1$.

With Z : $Z = 0.5e + 0.3e^2 + 0.2 = 3.776$, so $\pi^* = (0.360, 0.587, 0.053)$ and $\pi^*(y_2)/\pi^*(y_1) = 1.631$.

Without Z : $(0.3/0.5) e^{2-1} = 1.631$. **Same number: a ratio never needs Z .**

DPO trains on preference pairs, so it never needs Z

- **Data:** for one prompt, a person saw two responses and preferred y_w (winner) to y_l (loser).
- **Model of the choice** (Bradley–Terry): $P(y_w \text{ preferred}) = \sigma(r(y_w) - r(y_l))$, with $\sigma(z) = 1/(1 + e^{-z})$. Only the **difference** of rewards matters.

Take logarithms of the ratio on the previous slide and solve for that difference:

$$r(y_w) - r(y_l) = \beta \ln \frac{\pi^*(y_w)}{\pi_{\text{ref}}(y_w)} - \beta \ln \frac{\pi^*(y_l)}{\pi_{\text{ref}}(y_l)}.$$

Direct Preference Optimization (DPO). Replace π^* by the model π being trained, and maximize the average over preference pairs of

$$\ln \sigma \left(\beta \ln \frac{\pi(y_w)}{\pi_{\text{ref}}(y_w)} - \beta \ln \frac{\pi(y_l)}{\pi_{\text{ref}}(y_l)} \right).$$

No Z and no separate reward model: only ratios of probabilities appear. Rafailov et al., NeurIPS 2023.

Maximum entropy gives the form of the model, not its numbers

1

Maximum entropy

2

Moment matching

3

Monte Carlo

4

Markov chains

5

Detailed balance

6

Metropolis–Hastings

7

Gibbs sampling

8

Mixing

WHAT WE HAVE

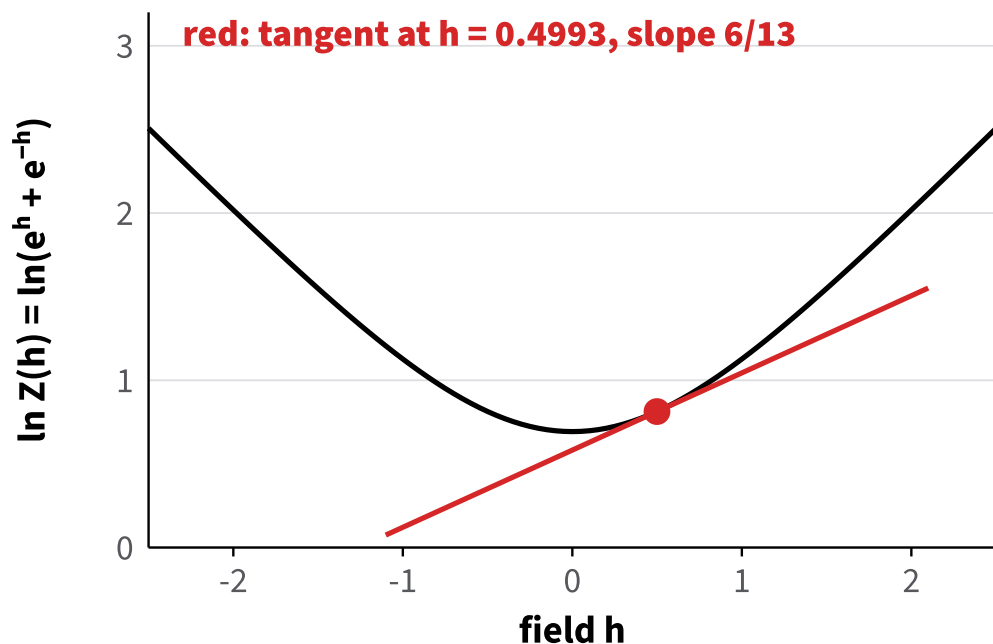
- The form $p_{\theta}(x) = \exp(\sum_k \theta_k \phi_k(x)) / Z(\theta)$.
- Each constrained average draws one factor.

WHAT IS MISSING

- The numbers θ . On the chain we quoted them; in general we must find them.
- We only have data: records $x^{(1)}, \dots, x^{(N)}$.

NEXT QUESTION Which θ reproduces the averages measured **2 • Moment matching** in the data?

The slope of $\ln Z$ is the model's average



The smallest case: one spin with field h ,
 $p_h(s) = e^{hs} / Z(h)$.

$$Z(h) = e^h + e^{-h},$$

$$\frac{d \ln Z}{dh} = \frac{e^h - e^{-h}}{e^h + e^{-h}} = \tanh h.$$

$$\mathbb{E}_h[s] = (+1) \frac{e^h}{Z} + (-1) \frac{e^{-h}}{Z} = \tanh h.$$

The same number: the slope of $\ln Z$ is the average of s .

- **Fit to data** $\mathbb{E}[s_1] = \frac{6}{13}$: solve $\tanh h = \frac{6}{13}$, so
 $h = \frac{1}{2} \ln \frac{19}{7} = 0.4993$ (red tangent).

The derivative of $\ln Z$ is each feature's model average

Write $A(\theta) = \ln Z(\theta) = \ln \sum_x e^{\sum_j \theta_j \phi_j(x)}$ and differentiate term by term:

$$\frac{\partial A}{\partial \theta_k} = \frac{1}{Z(\theta)} \sum_x \phi_k(x) e^{\sum_j \theta_j \phi_j(x)} = \sum_x \phi_k(x) p_\theta(x) = \mathbb{E}_\theta[\phi_k].$$

For spins at temperature T . $p_T \propto e^{-E/T}$ with $E(s) = -\sum_i h_i s_i - \sum_{i < j} J_{ij} s_i s_j$, so the parameters enter as h_i/T and J_{ij}/T :

$$\frac{\partial \ln Z_T}{\partial h_i} = \frac{\mathbb{E}[s_i]}{T}, \quad \frac{\partial \ln Z_T}{\partial J_{ij}} = \frac{\mathbb{E}[s_i s_j]}{T}.$$

Maximum likelihood moves each parameter by data average minus model average

Setting. Records $x^{(1)}, \dots, x^{(N)}$; data averages $m_k = \frac{1}{N} \sum_n \phi_k(x^{(n)})$. Log-likelihood per record:

$$\ell(\theta) = \frac{1}{N} \sum_{n=1}^N \ln p_{\theta}(x^{(n)}) = \sum_k \theta_k m_k - \ln Z(\theta).$$

Theorem 4 (moment matching).

$$\frac{\partial \ell}{\partial \theta_k} = \underbrace{m_k}_{\text{data average}} - \underbrace{\mathbb{E}_{\theta}[\phi_k]}_{\text{model average}}.$$

At a maximum every model average equals its data average: $\mathbb{E}_{\theta}[\phi_k] = m_k$ for every k .

These are the same equations that appeared in Theorem 2. The next slide shows why that matters.

Maximum likelihood and maximum entropy give the same model

Recall Theorem 2. Among **all** distributions q with the averages $\mathbb{E}_q[\phi_k] = m_k$, the largest entropy belongs to $p_\theta(x) = \exp(\sum_k \theta_k \phi_k(x)) / Z(\theta)$ with the θ that gives those averages.

	searches over	picks	its answer satisfies
Maximum likelihood	the family p_θ only: K numbers θ_k	the θ that makes the data most probable	$\mathbb{E}_\theta[\phi_k] = m_k$ (Theorem 4)
Maximum entropy	every q with averages m_k	the most spread-out q	$q = p_\theta$ with $\mathbb{E}_\theta[\phi_k] = m_k$ (Theorem 2)

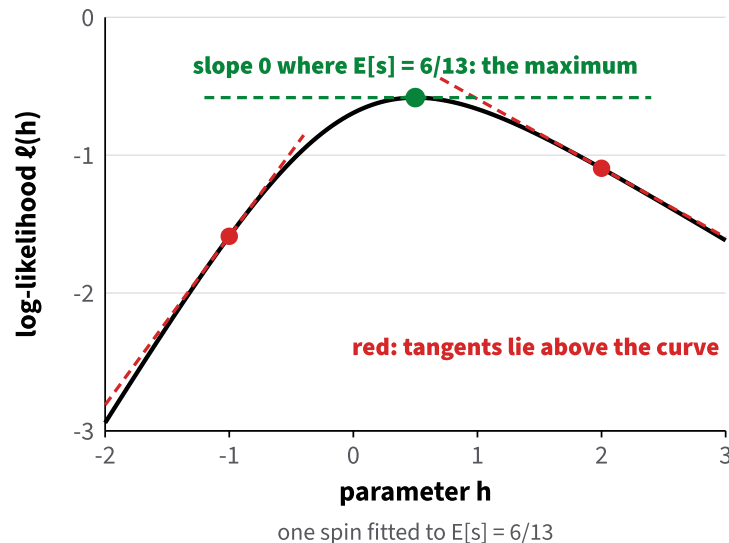
Same condition, same model. Fitting the family to data automatically gives the most spread-out distribution (largest entropy) with the data's averages: it assumes nothing beyond them.

On our chain. Fitting (animated shortly) ends at $h = (0.5756, -0.3922, 0.0719)$, $J_{12} = 0.2291$, $J_{23} = 0.6212$, the maximum-entropy numbers: both return the chain.

The log-likelihood is concave, so matching moments finds the global maximum

Recall $A(\theta) = \ln Z(\theta) = \ln \sum_x \exp \left(\sum_k \theta_k \phi_k(x) \right)$; its first derivatives are the model averages $\mathbb{E}_\theta[\phi_k]$. Differentiating once more:

$$\frac{\partial^2 A}{\partial \theta_k \partial \theta_l} = \mathbb{E}_\theta[\phi_k \phi_l] - \mathbb{E}_\theta[\phi_k] \mathbb{E}_\theta[\phi_l] = \text{Cov}_\theta(\phi_k, \phi_l).$$



- A covariance matrix is positive semidefinite: $v^\top \text{Cov}(\phi) v = \text{Var} \left(\sum_k v_k \phi_k \right) \geq 0$. So A is convex and $\ell(\theta) = \theta \cdot m - A(\theta)$ is **concave**.
- A concave function lies below each of its tangents: $\ell(\theta') \leq \ell(\theta) + \nabla \ell(\theta) \cdot (\theta' - \theta)$ for all θ' .
- Where all moments match, $\nabla \ell(\theta) = 0$ (Theorem 4), so $\ell(\theta') \leq \ell(\theta)$ for **every** θ' . **A global maximum; gradient ascent has no false peak to get stuck on.**

Does a matching θ exist, and is it unique?

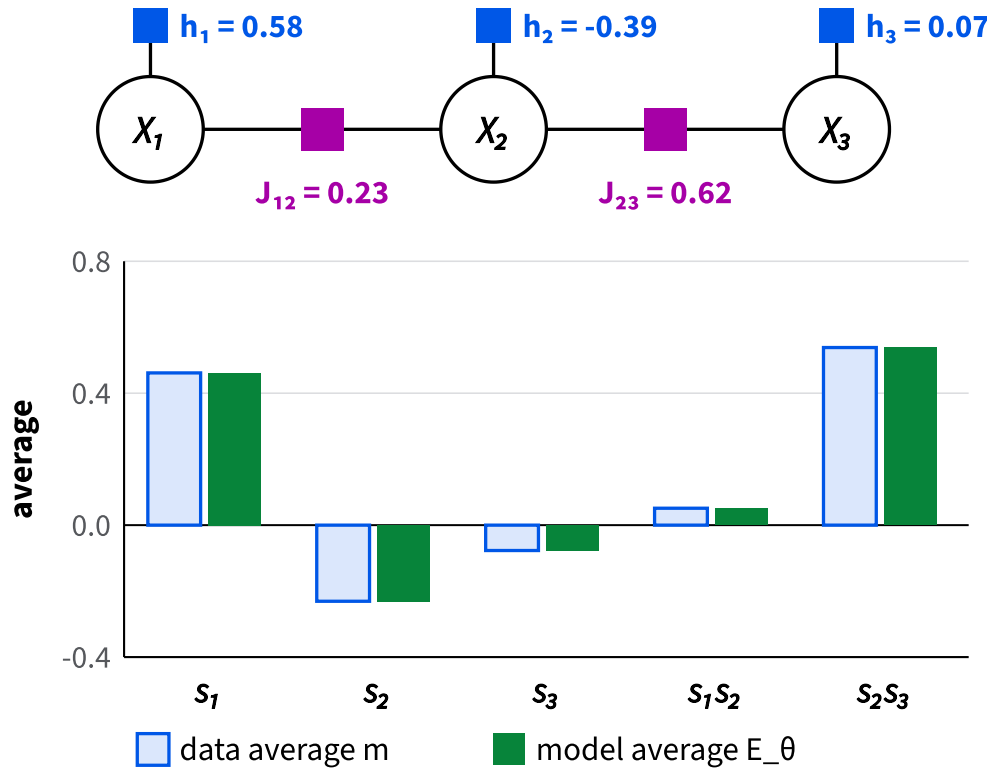
Existence. The argument above needs a θ whose averages equal m . For an extreme average such as $\mathbb{E}[s_1] = 1$, no finite θ works (next slides).

Uniqueness and linear dependence. The features are **linearly dependent** if some $v \neq 0$ makes $\sum_k v_k \phi_k(x)$ the **same constant for every** x . Then $\text{Var}(\sum_k v_k \phi_k) = 0$: the log-likelihood is flat along v , and many θ fit equally well.

Example: one binary x with $\phi_1 = x$, $\phi_2 = 1 - x$. Since $\phi_1 + \phi_2 = 1$, the parameters θ and $\theta + t(1, 1)$ multiply every weight by the same e^t , which cancels in Z : the same model.

With linearly independent features, θ is unique. Either way, the fitted **distribution** is unique (Theorem 2); only its parameters can be redundant.

Fitting turns five knobs until the model's averages equal the data's



- Start: every knob at 0. The model is uniform and every average is 0.
- Rule:** each knob moves by *data average* – *model average* of its own feature:
- $\theta_k \leftarrow \theta_k + \eta (m_k - \mathbb{E}_\theta[\phi_k]), \eta = 0.5.$
- The knobs settle at our chain's h and J .
- Log-likelihood: from $-\ln 8 = -2.0794$ up to $-H(p) = -1.7698$.

step **40** · log-likelihood per record **-1.7698** (maximum $-H(p) = -1.7698$)

The first fitting step on our chain, by hand

Data averages (the five averages of the chain, as before):

$$m = (\mathbb{E}[s_1], \mathbb{E}[s_2], \mathbb{E}[s_3], \mathbb{E}[s_1 s_2], \mathbb{E}[s_2 s_3]) = (0.4615, -0.2308, -0.0769, 0.0513, 0.5385).$$

Step 0. All knobs are 0, so the model is uniform on the 8 states: each spin is ± 1 with probability $\frac{1}{2}$, and **every model average is 0**. The gradient is $m - 0 = m$.

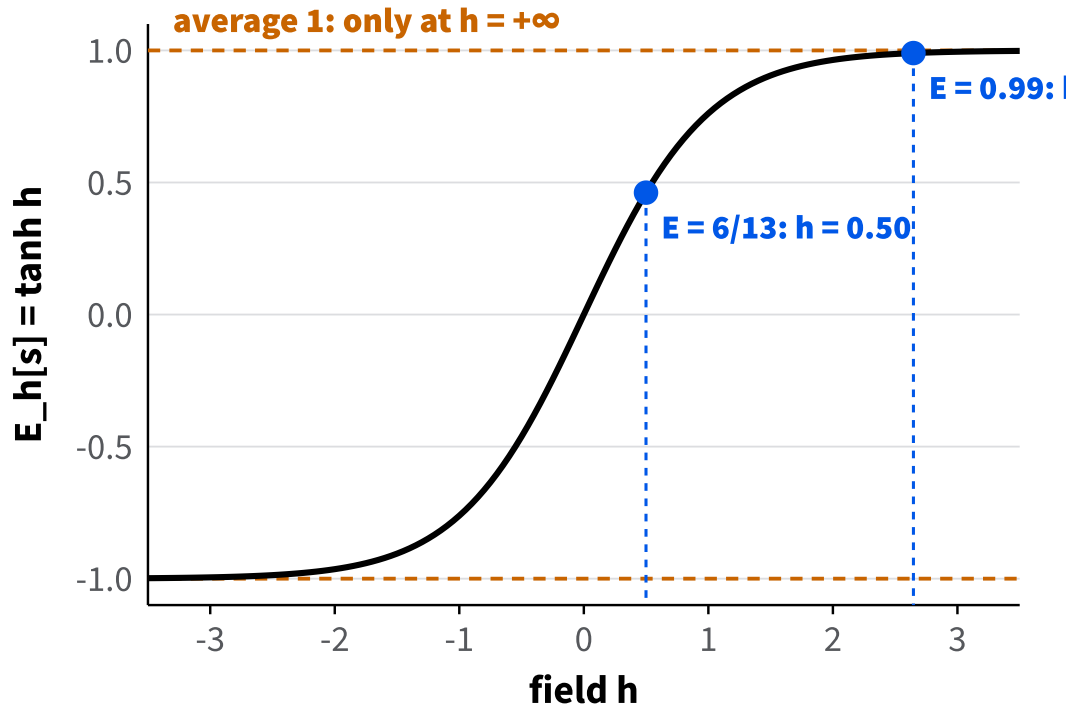
Step 1. With $\eta = 0.5$, every knob moves by half its data average:

$$(h_1, h_2, h_3, J_{12}, J_{23}) = 0.5 m = (0.2308, -0.1154, -0.0385, 0.0256, 0.2692).$$

J_{23} moves most because $s_2 s_3$ has the largest data average.

Knobs interact. With the fields alone, the spins are independent and $\mathbb{E}[s_i] = \tanh h_i$ (the one-spin slide), so $\mathbb{E}[s_1 s_2] \approx \tanh(0.2308) \tanh(-0.1154) = 0.2268 \times (-0.1149) = -0.026$. The small $J_{12} = 0.0256$ pulls it back only to -0.003 , below zero, although the data say $+0.0513$. **Each knob changes the other averages, so fitting needs many small steps.**

An extreme average needs an infinite parameter



- One spin: $\mathbb{E}_h[s] = \tanh h$ (the one-spin slide). The curve approaches ± 1 but never reaches it.
- Data average $\frac{6}{13}$: $h = 0.4993$. Data average 0.99: $h = 2.65$.
- If $X_1 = 1$ in **every** record, the data average is exactly 1, and no finite h gives $\tanh h = 1$: the fit runs off to $h = \infty$.
- XOR's missing rule has the same problem (next slide). In practice, add a few made-up records that smooth the counts, so no average is exactly ± 1 .

Test the fitted model on an average you did not fit

Fitted averages match **by construction**, so they cannot reveal a wrong model. Only an average left out of the fit can.

	chain data (our chain)	XOR data
averages fitted	$s_1, s_2, s_3, s_1s_2, s_2s_3$	s_x, s_y, s_w and all pairs
average left out	s_1s_3	$s_xs_ys_w$
model / data	0.0513 / 0.0513: pass	0 / -1: fail

Chain. $x_1 = x_3$ in 000, 010, 101, 111 (weight 41), otherwise 37: $\mathbb{E}[s_1s_3] = \frac{41-37}{78} = 0.0513$.

XOR. $s_xs_ys_w = -1$ in every data state 000, 011, 101, 110; the uniform pair fit has +1 on 4 states and -1 on 4, so 0. **A three-variable factor is missing.**

Next-token training: the gradient of one record

Fix one context c , say “The cat sat on the”. The network reads c and outputs one real number per vocabulary token k , its **logit** z_k . (In a transformer, $z_k = w_k \cdot h_c$: the final hidden state h_c times token k 's output vector w_k .) The softmax turns logits into probabilities:

$$p_\theta(k) = \frac{e^{z_k}}{\sum_j e^{z_j}} : \text{the exponential family with } \phi_k = 1[\text{next token} = k] \text{ and knob } \theta_k = z_k.$$

One record: context c , then token y . Its loss is the surprise of y :

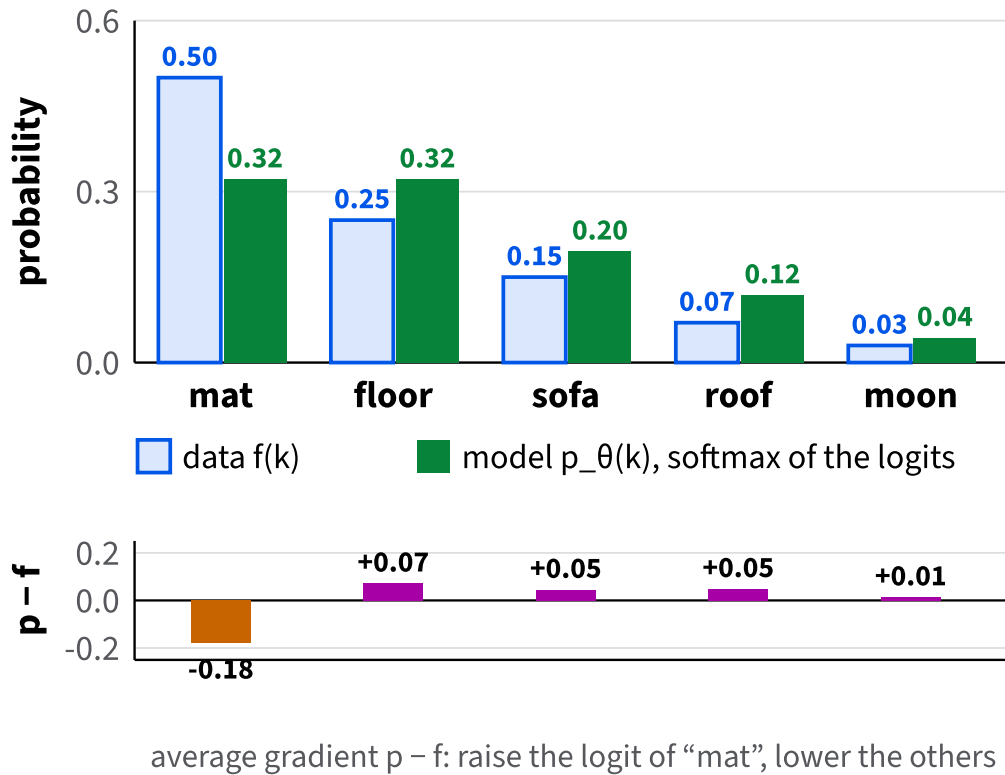
$$-\ln p_\theta(y) = -z_y + \ln \sum_j e^{z_j},$$

$$\frac{\partial(-\ln p_\theta(y))}{\partial z_k} = -1[k = y] + \frac{e^{z_k}}{\sum_j e^{z_j}} = p_\theta(k) - 1[k = y].$$

One record alone is a poor target: $1[k = y]$ is 0 or 1, extreme (Warning 1). Next: average over the corpus.

Over the corpus, next-token training is moment matching

After “The cat sat on the” (illustrative numbers)



- The context c occurs n times in the corpus, followed by tokens $y^{(1)}, \dots, y^{(n)}$.

- Average the one-record gradient over them:

$$\frac{1}{n} \sum_{i=1}^n (p_\theta(k) - 1[y^{(i)} = k]) = p_\theta(k) - f(k),$$

where $f(k)$ is the **fraction of those records whose next token is k** : the data average of ϕ_k .

- Zero gradient $\iff p_\theta(k) = f(k)$: the model's probabilities match the corpus frequencies.
- **This is Theorem 4:** the log-likelihood gradient is data average minus model average, $m_k - \mathbb{E}_\theta[\phi_k]$; here $m_k = f(k)$ and $\mathbb{E}_\theta[\phi_k] = p_\theta(k)$. The loss is minus the log-likelihood, so its gradient is $p_\theta(k) - f(k)$.

Boltzmann machines learn by data correlation minus model correlation

$$J_{ij} \leftarrow J_{ij} + \underbrace{\eta \left(\langle s_i s_j \rangle_{\text{data}} - \langle s_i s_j \rangle_{\text{model}} \right)}_{\Delta J_{ij}, \text{ the change in one step}} \quad (\text{Ackley, Hinton and Sejnowski, 1985})$$

- $\langle \cdot \rangle$ means an average: $\langle s_i s_j \rangle_{\text{data}}$ over the training examples, $\langle s_i s_j \rangle_{\text{model}} = \mathbb{E}_{\theta}[s_i s_j]$. This is our fitting rule for the knob J_{ij} (Theorem 4).
- Data term: cheap. Model term: a sum over all 2^n spin states, estimated by **running the Boltzmann machine**, that is, by sampling.
- **Contrastive divergence** starts that sampler at the data and runs only a few steps (Hinton, 2002). Energy-based image models do the same (Du and Mordatch, 2019).

Ackley, Hinton, Sejnowski, *Cognitive Science* 1985; Hinton, *Neural Computation* 2002; Du, Mordatch, *NeurIPS* 2019. Full references at the end.

The data average is easy; the model average is the impossible sum

1
Maximum
entropy

2
Moment
matching

3
Monte Carlo

4
Markov chains

5
Detailed
balance

6
Metropolis–
Hastings

7
Gibbs sampling

8
Mixing

WHAT WE HAVE

- A fitting rule: $\theta_k \leftarrow \theta_k + \eta (m_k - \mathbb{E}_{\theta}[\phi_k])$.
- The data average m_k : a sum over N records.

WHAT BLOCKS US

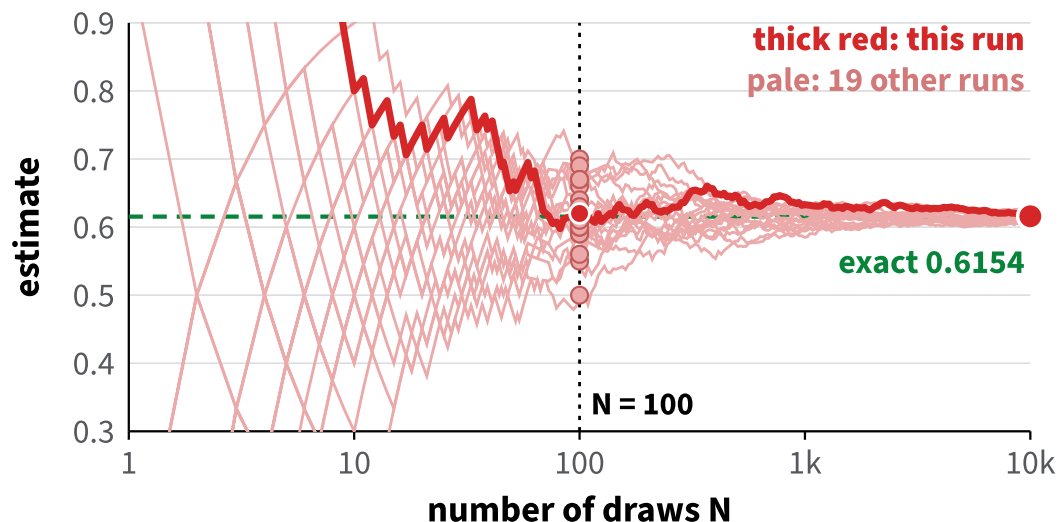
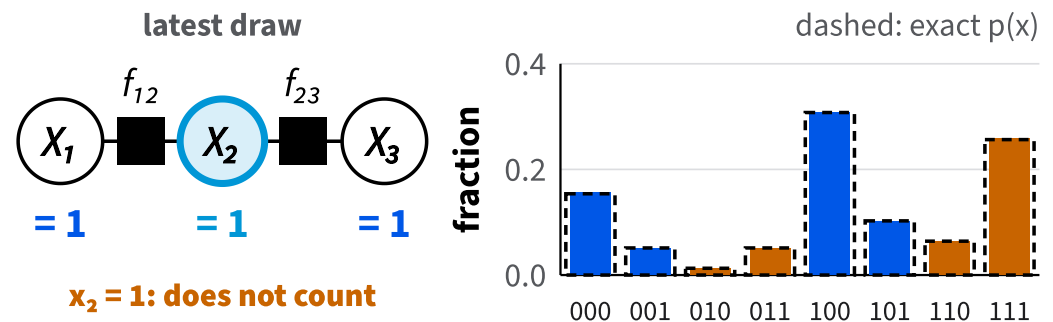
- The model average $\mathbb{E}_{\theta}[\phi_k] = \sum_x p_{\theta}(x) \phi_k(x)$: a sum over every state, and it needs Z .
- Chain: 8 terms. 32×32 grid: 2^{1024} terms, at every fitting step.

NEXT QUESTION Can we estimate an average without computing the sum? **3 • Monte Carlo**

Ten minutes

After the break: an average over 10^{308} states, estimated from a few thousand draws, and why the error does not care how many states there are.

Draw states, count, divide: an average without the sum



$N = 10000$ · estimate **0.6159** · distance from 0.6154: **0.0005**

- **Top:** a draw sets **every variable** (left). Bars: **fraction of draws** per state; dashed: exact $p(x)$. Blue bars ($x_2 = 0$) **sum to the estimate**.
- **Bottom:** the **estimate** as the number of draws N grows, for this run (**thick red line**). Green dashed line: the **exact** $P(X_2 = 0) = 48/78 = 0.6154$.

Repeat the experiment with $N = 100$ draws.
How far from 0.6154 is the estimate typically?

0.5

0.05

0.005

Answer. About **0.05**. Pale lines: 19 more runs. At the dotted line $N = 100$ the 20 estimates $\hat{\mu}^{(k)}$ lie **between 0.50 and 0.70**, and

$$\sqrt{\frac{1}{20} \sum_k (\hat{\mu}^{(k)} - 0.6154)^2} = 0.047.$$

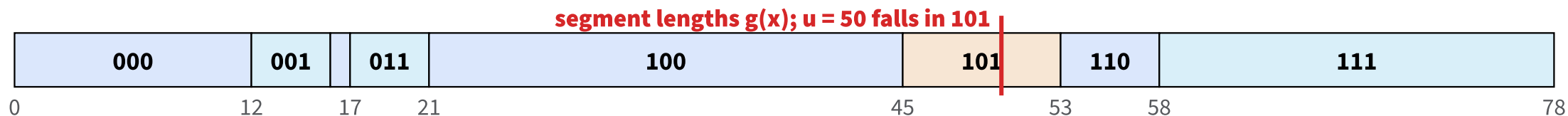
Derived exactly on the next slides: **0.0487**.

Counting is averaging: every probability is the average of an indicator

$$P(X_2 = 0) = \sum_x p(x) 1[x_2 = 0] = \mathbb{E}_p[1[X_2 = 0]].$$

- So "the fraction of draws with $x_2 = 0$ " is a **sample average** of the **indicator** $g(x) = 1[x_2 = 0]$.
- The same recipe works for **any quantity**: $\mathbb{E}[s_1 s_2]$, $\mathbb{E}[E]$, a marginal, a model average needed for fitting.
- **Law of large numbers**: for independent draws from p , the sample average **converges** to $\mathbb{E}_p[g]$ as $N \rightarrow \infty$.
- **Open**: the 20 runs spread less as N grows. **How fast** does the spread shrink? That decides whether the method is useful.

How one independent draw from p is made on our chain



Lay the eight weights side by side on $[0, 78)$. A uniform random u lands in one segment, so state x is hit with probability $g(x)/78 = p(x)$. Above: $u = 50$ lands in 101.

```
cum = [12, 16, 17, 21, 45, 53, 58, 78]    # running totals of g over 000, 001, ..., 111
u = 78 * random.random()                    # uniform on [0, 78)
x = states[bisect.bisect_right(cum, u)]     # the state whose segment contains u
```

Each draw uses fresh randomness, so the draws are **independent**. But the recipe needs the total $Z = 78$: fine for 8 states, impossible for 2^{1024} .

Monte Carlo is unbiased, and its error shrinks like $1/\sqrt{N}$

Definition (Monte Carlo estimate). Independent draws $x^{(1)}, \dots, x^{(N)}$ from p :

$$\hat{\mu}_N = \frac{1}{N} \sum_{t=1}^N g(x^{(t)}) \approx \mu = \mathbb{E}_p[g] = \sum_x p(x) g(x).$$

Theorem 5. $\mathbb{E}[\hat{\mu}_N] = \mu$ (**unbiased**) and $\text{Var}(\hat{\mu}_N) = \sigma^2/N$, where $\sigma^2 = \text{Var}_p(g) = \mathbb{E}_p[(g(X) - \mu)^2]$ is the **variance of one draw**.

Proof. Expectation is **linear**: $\mathbb{E}[\hat{\mu}_N] = \frac{1}{N} \sum_t \mathbb{E}[g(x^{(t)})] = \frac{1}{N} \cdot N\mu$. The draws are independent, so their **variances add**: $\text{Var}(\hat{\mu}_N) = \frac{1}{N^2} \cdot N\sigma^2$. ■

Definition (standard error). $\text{SE} = \sqrt{\text{Var}(\hat{\mu}_N)} = \sigma/\sqrt{N}$. Since $\mathbb{E}[\hat{\mu}_N] = \mu$, it equals the root-mean-square error $\sqrt{\mathbb{E}[(\hat{\mu}_N - \mu)^2]}$: **the typical distance of the estimate from μ** .

The proof uses only linearity and independence, no central limit theorem. **A. B. Owen.** *Monte Carlo Theory, Methods and Examples*. 2013, Chapter 2.

The standard error on the chain, seen in the runs

One draw scores $g = 1[x_2 = 0]$: **1 with probability** $\mu = 48/78 = 0.6154$, and **0 with probability** $30/78 = 0.3846$. Since $g^2 = g$,

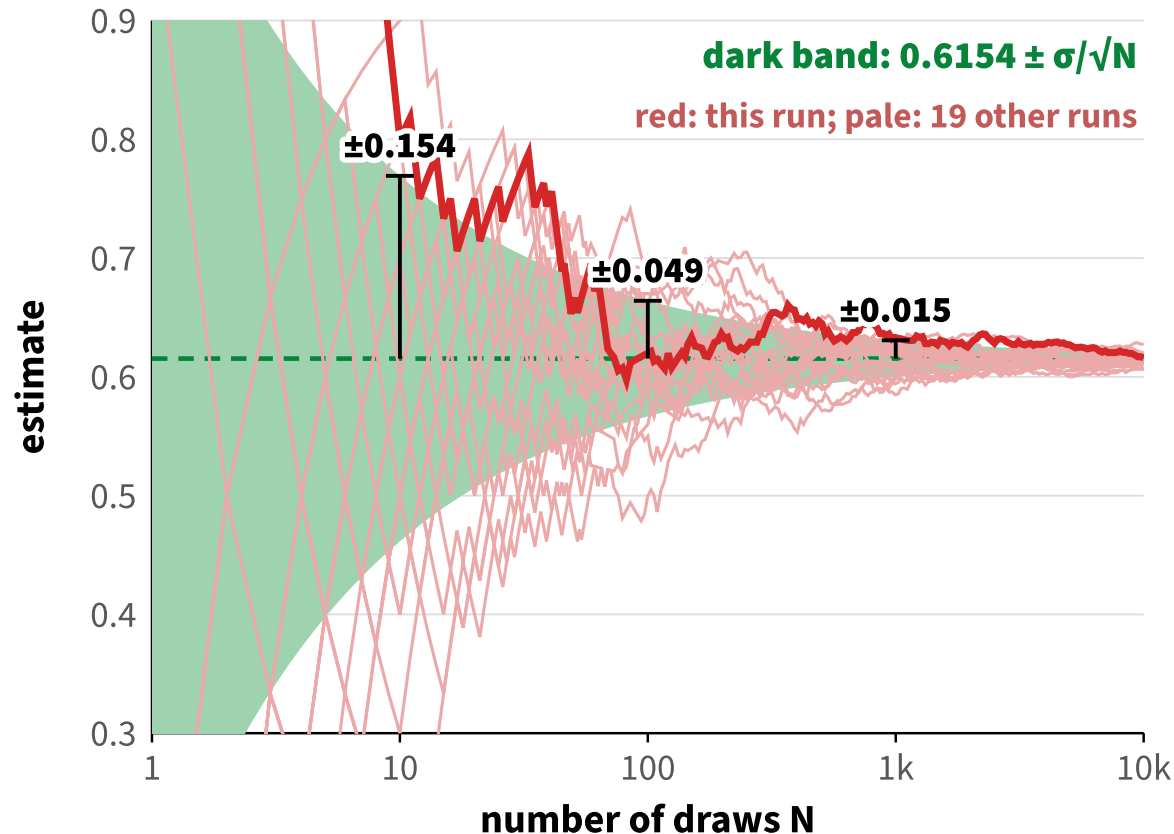
$$\sigma^2 = \mathbb{E}_p[g^2] - \mu^2 = \mu - \mu^2 = \mu(1 - \mu),$$

$$\sigma = \sqrt{0.6154 \times 0.3846} = 0.4865.$$

So $SE = 0.4865/\sqrt{N}$, and at $N = 100$, **SE = 0.0487**. The 20 runs on the right measured **0.047**.

Dark band: $0.6154 \pm SE$ at each N . The runs wander **in and out** of it; its half-width is marked at $N = 10, 100, 1000$.

In practice μ , and so σ , is unknown: **plug in the estimate**, $\hat{\sigma} = \sqrt{\hat{\mu}_N(1 - \hat{\mu}_N)}$.



On the chain: a hundred times more draws buys one more correct digit

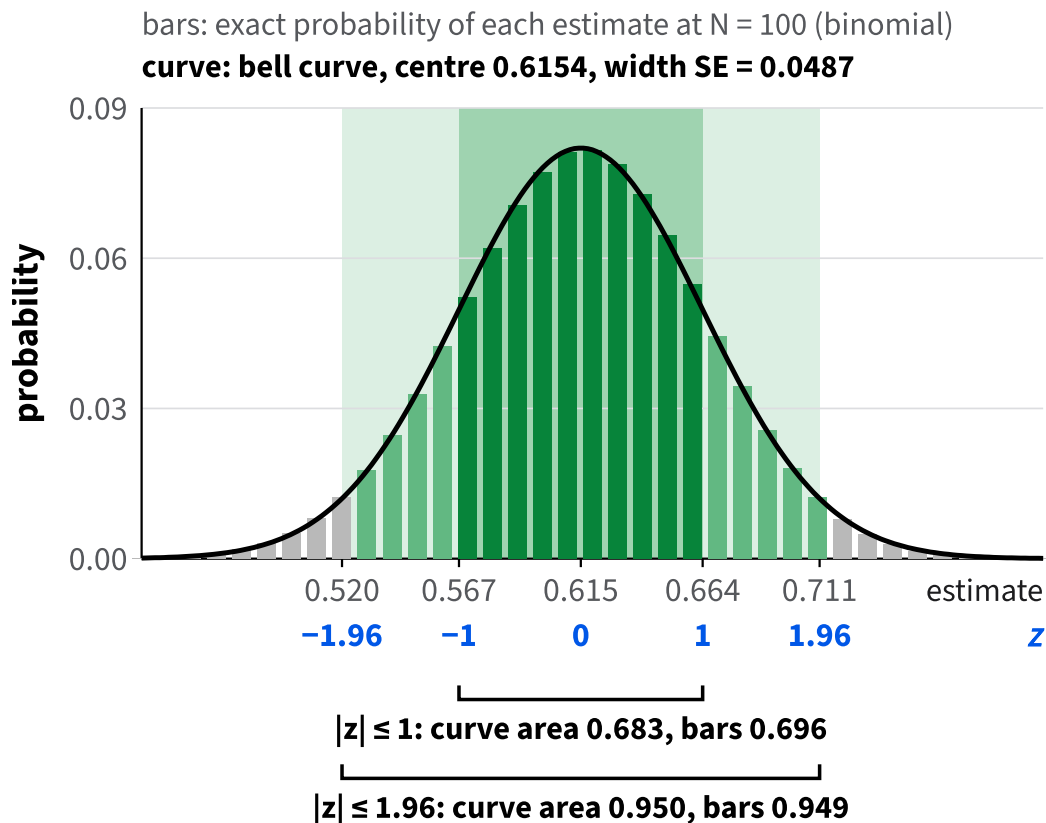
$g = 1[x_2 = 0], \mu = 0.6154:$

$$\sigma = \sqrt{\mu(1 - \mu)} = 0.4865.$$

N	standard error $\sigma/\sqrt{N}$
10	0.1538
100	0.0487
1,000	0.0154
10,000	0.0049

- **The number of states appears nowhere.** The error depends only on N and σ .
- A 32×32 grid, 2^{1024} states: **the same N gives the same error**, if σ is similar.
- **The price:** the error falls slowly. **Halving it needs four times as many draws.**

The central limit theorem gives the shape of the spread



Central limit theorem. Independent draws with finite $\sigma > 0$. For every z , as $N \rightarrow \infty$,

$$P\left(\frac{\hat{\mu}_N - \mu}{\text{SE}} \leq z\right) \rightarrow P(Z \leq z),$$

where Z is a **standard normal** variable, whose density is the **bell curve** $e^{-z^2/2}/\sqrt{2\pi}$.

The ratio counts **standard errors from μ** (blue row, left). Under the bell curve, $P(|Z| \leq 1) = 0.683$ and $P(|Z| \leq 1.96) = 0.950$, so for large N

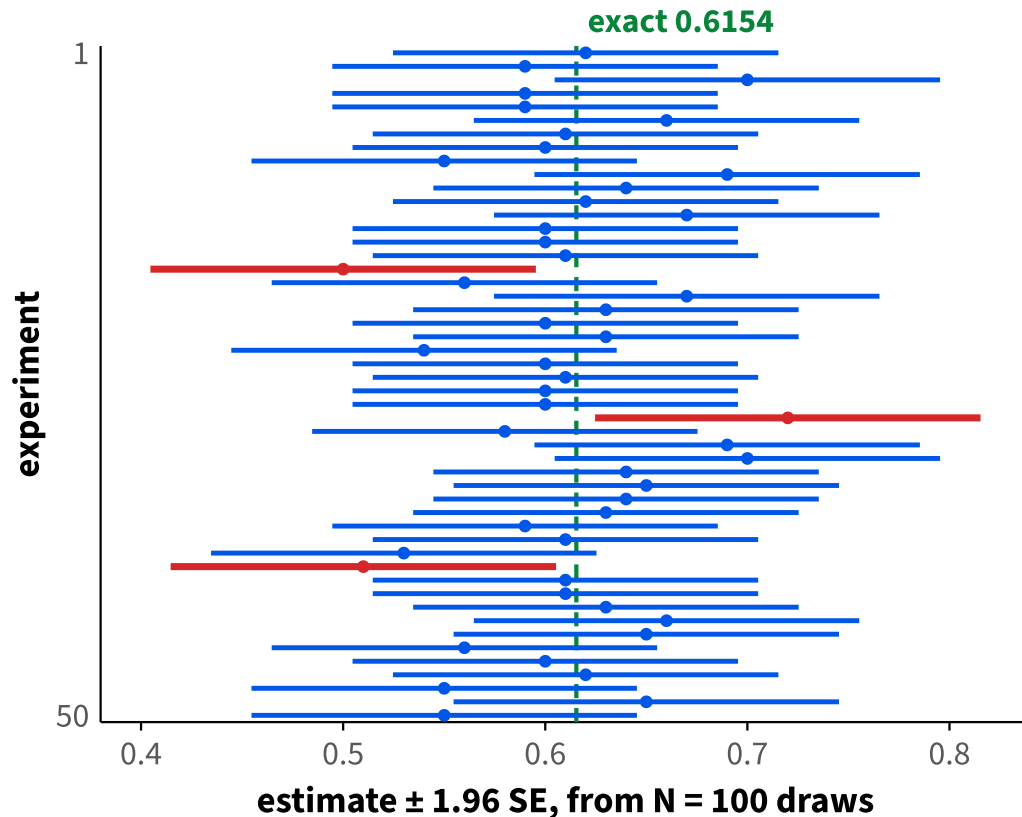
$$P(|\hat{\mu}_N - \mu| \leq \text{SE}) \approx 0.683,$$

$$P(|\hat{\mu}_N - \mu| \leq 1.96 \text{ SE}) \approx 0.950.$$

Check: `norm.cdf(1) - norm.cdf(-1)` in SciPy. **1.96 is chosen to make the area 0.95.**

L. Wasserman. *All of Statistics*. Springer, 2004, Chapter 5.

A 95% interval: right in 95% of repeated experiments



47 of 50 intervals contain 0.6154

Definition (95% interval). From one experiment of N draws, report

$$\hat{\mu}_N \pm 1.96 \text{ SE}, \quad \text{SE} = \sigma / \sqrt{N}.$$

It contains μ exactly when $|\hat{\mu}_N - \mu| \leq 1.96 \text{ SE}$: **in 94.9% of experiments** at $N = 100$. Left: **47 of 50** intervals contain 0.6154; 3 (red) miss.

Experiment 1 gives $0.62 \pm 0.095 = [0.525, 0.715]$. Is 0.6154 in it with probability 0.95?

Yes

No

Answer. No. Once drawn, nothing is random: 0.6154 is in $[0.525, 0.715]$, full stop. **The 95% belongs to the recipe:** over many experiments, 95% of its intervals contain μ .

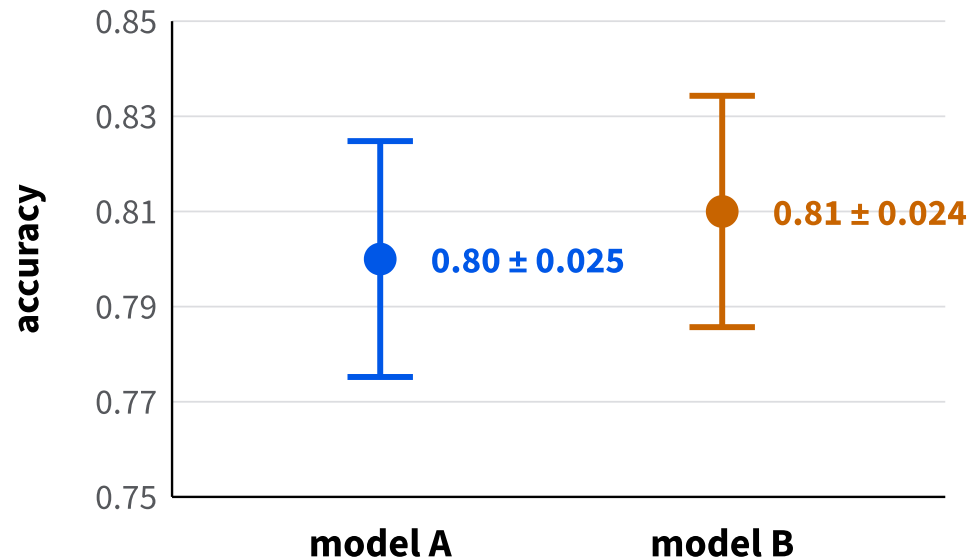
All five model averages that fitting needs, from 1,000 draws

average	exact	estimate, N = 1000	standard error	error / SE
$E[s_1]$	0.4615	0.4820	0.0281	0.73
$E[s_2]$	-0.2308	-0.2320	0.0308	0.04
$E[s_3]$	-0.0769	-0.0880	0.0315	0.35
$E[s_1 s_2]$	0.0513	0.0500	0.0316	0.04
$E[s_2 s_3]$	0.5385	0.5400	0.0266	0.06

One run of 1,000 direct draws from p , with a fixed seed. The model here is p itself, so the exact values are known.

- Fitting needs the **model averages** $\mathbb{E}_\theta[\phi_k]$, each a sum over all states. **One set of draws estimates all five.**
- Each ϕ_k is ± 1 , so $\sigma^2 = 1 - m^2$ and $\text{SE} = \sqrt{1 - m^2} / \sqrt{1000} \approx 0.03$.
- Last column: **error ÷ SE**. All below 1.96, so **every 95% interval contains the exact value.**

Every benchmark score is a Monte Carlo estimate, and needs an error bar

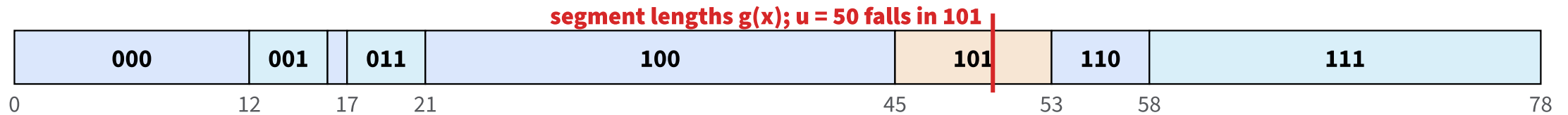
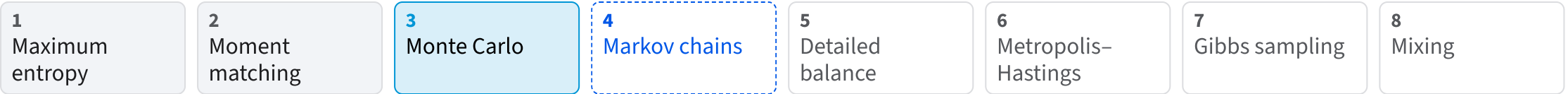


$N = 1,000$ questions; bars: 95% intervals

- Accuracy a on N questions: the **average of N draws of a 0/1 score**, like $1[x_2 = 0]$. So $SE = \sqrt{a(1-a)/N}$.
- Model A, $a = 0.80$, $N = 1000$: $SE = 0.0126$, **95% interval** 0.80 ± 0.025 .
- **Is model B, at 0.81, better?** The gap 0.01 has $SE = \sqrt{0.0126^2 + 0.0124^2} = 0.018$: 95% interval 0.01 ± 0.035 , which **contains 0. Not yet a real difference.**

Comparing both models question by question shrinks the gap's SE (Miller, 2024). $\text{pass}@k$ is estimated the same way (Chen et al., 2021). **E. Miller.** Adding error bars to evals. arXiv:2411.00640, 2024. **M. Chen et al.** Evaluating large language models trained on code. arXiv:2107.03374, 2021.

Drawing directly from p needs Z



WHAT BLOCKS US

- Direct drawing: pick u in $[0, 78)$ and find its segment. That needs **every weight and their total Z** .

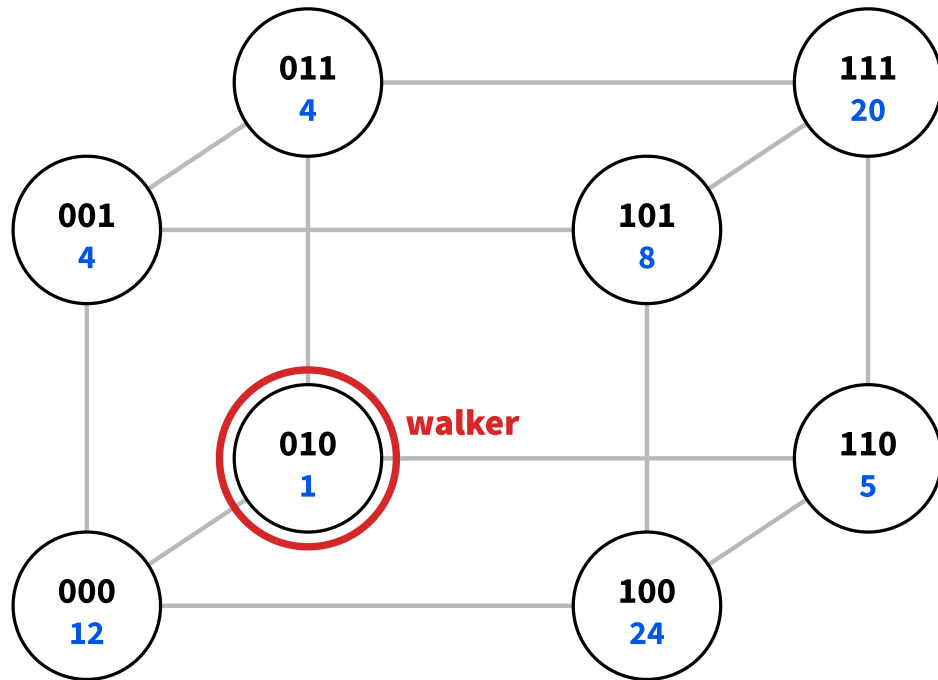
WHAT IS CHEAP

- **Ratios:** $\frac{p(x')}{p(x)} = \frac{g(x')}{g(x)}$, so Z **cancels**.
- **Neighbours:** $\frac{g(100)}{g(000)} = \frac{f_{12}(1,0)}{f_{12}(0,0)} = \frac{8}{4} = 2$; even f_{23} cancels.

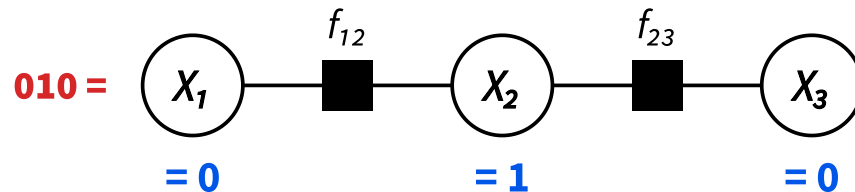
NEXT QUESTION Can a walk that only compares neighbouring states produce draws from p ?

4 • Markov chains

The eight states form a map: neighbours differ in one variable

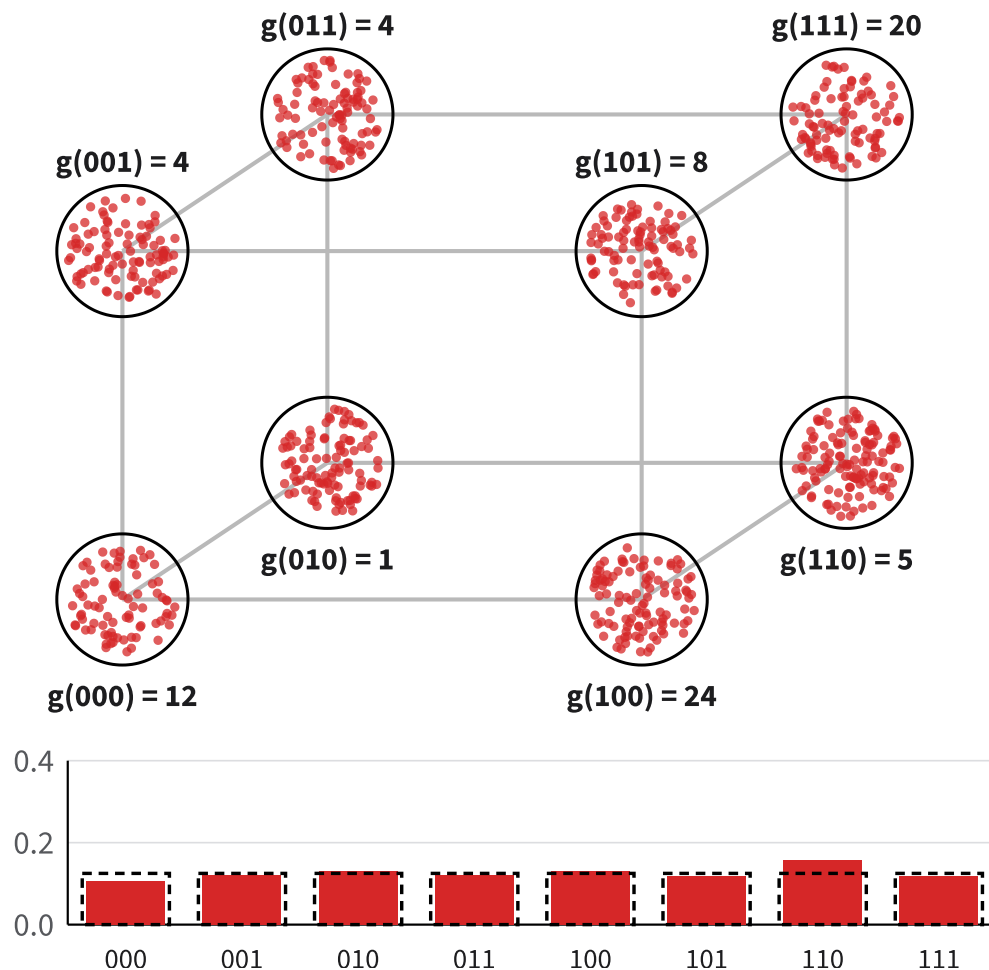


inside each state: its weight $g(x)$



- A **state** is one assignment of the chain, $x = (x_1, x_2, x_3)$. Inside each circle: its **weight** $g(x) = f_{12}(x_1, x_2) f_{23}(x_2, x_3)$.
- Two states are joined when they **differ in one variable**: flip X_1 , X_2 or X_3 . Every state has **three neighbours**.
- A **walker** stands on a state and moves along edges. Below the cube: its state 010 written on the chain.
- A move needs only the current state and its neighbours: **no Z**.
- **Question:** release many walkers moving at random. Where do they end up?

Release 800 walkers: they forget their start and settle



step **30** · distance of the walkers to uniform ($1/8$ each): **0.043**

- **Start:** all 800 walkers at 010.
- **Each step, each walker** picks one of the three variables at random and sets it by a **fair coin**, so it may keep its value.
- **Distance** to uniform: $\frac{1}{2} \sum_x |\mu(x) - \frac{1}{8}|$, with $\mu(x)$ the fraction of walkers at x : **the fraction of walkers that would have to move**. Exact: 0.875, 0.500, 0.333, 0.222, ... $\rightarrow 0$.
- **But the walkers settle at $\frac{1}{8}$ per state, not at $p(x) = g(x)/78$.** The weights on the cube are $g(x) = f_{12}(x_1, x_2)f_{23}(x_2, x_3)$: $g(010) = 1 \cdot 1 = 1$ and $g(100) = 8 \cdot 3 = 24$, so $p(010) = 1/78$ and $p(100) = 24/78$.

A Markov chain chooses the next state by looking only at the current one

Definition (Markov chain). States $X_0, X_1, X_2, \dots$ with a **transition matrix**

$$P(x, x') = \Pr(X_{t+1} = x' \mid X_t = x), \quad P(x, x') \geq 0, \quad \sum_{x'} P(x, x') = 1 \text{ (each row).}$$

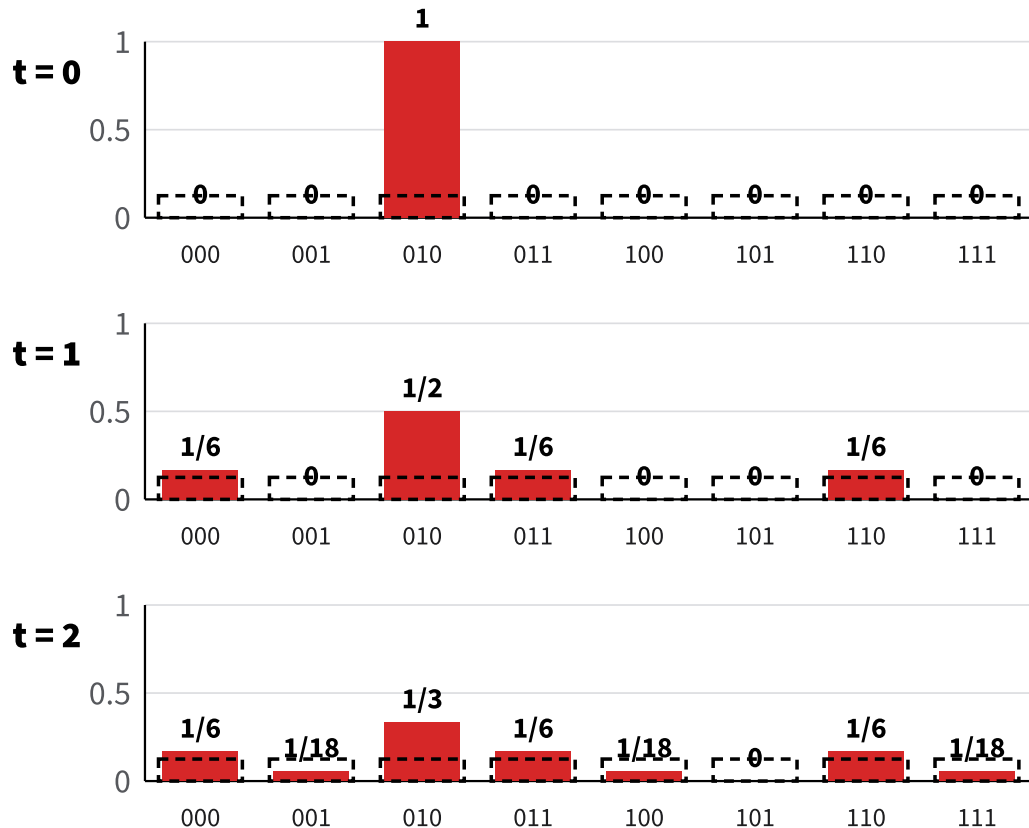
Given the present, **the past carries no extra information** about the next step.

The fair-coin walk, row of state 000. Pick X_1, X_2 or X_3 with probability $\frac{1}{3}$; the coin keeps the current value with probability $\frac{1}{2}$:

$$P(000, 000) = \frac{1}{2}, \quad P(000, 100) = P(000, 010) = P(000, 001) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}.$$

Row sum: $\frac{1}{2} + 3 \cdot \frac{1}{6} = 1$. Every row looks the same: **stay with $\frac{1}{2}$, go to each neighbour with $\frac{1}{6}$.**

The walkers' distribution moves by one matrix product per step



$\mu_t(x)$: the **fraction of walkers** at state x after t steps.
To be at x' next, a walker was at some x and moved:

$$\mu_{t+1}(x') = \sum_x \mu_t(x) P(x, x'), \quad \text{that is} \quad \mu_{t+1} = \mu_t P.$$

Fair-coin walk: each state **keeps half its walkers** and **receives a sixth of each neighbour's**:

$$\mu_{t+1}(x') = \frac{1}{2} \mu_t(x') + \frac{1}{6} \sum_{x \text{ next to } x'} \mu_t(x).$$

Left, from all at 010. Two entries of μ_2 :

$$\begin{aligned} \mu_2(010) &= \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{6} \left(\frac{1}{6} + \frac{1}{6} + \frac{1}{6} \right) = \frac{1}{3}, \\ \mu_2(100) &= \frac{1}{2} \cdot 0 + \frac{1}{6} \left(\frac{1}{6} + \frac{1}{6} + 0 \right) = \frac{1}{18}. \end{aligned}$$

A stationary distribution is one that a step leaves unchanged

Definition (stationary distribution). A distribution π with $\pi P = \pi$, that is $\sum_x \pi(x) P(x, x') = \pi(x')$ for every x' .

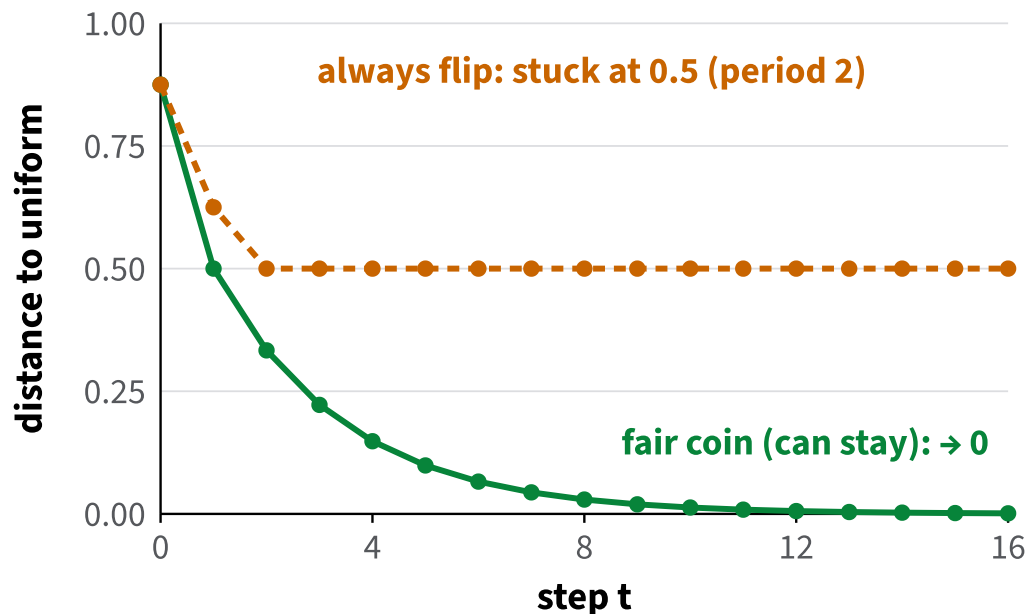
The fair-coin walk keeps the uniform $u = (\frac{1}{8}, \dots, \frac{1}{8})$. Its moves are **symmetric**, $P(x, x') = P(x', x)$: from 000 to 100 and back, $\frac{1}{6}$ each. So

$$\sum_x \frac{1}{8} P(x, x') = \frac{1}{8} \sum_x P(x', x) = \frac{1}{8} \cdot 1 = \frac{1}{8}.$$

This is **why the walkers went to** $\frac{1}{8}$ per state. To sample from p , **we need a chain whose stationary distribution is** p .

How fast. The animation's distance to u : 0.875, 0.500, 0.333, 0.222, 0.148. After step 1, **each step multiplies it by** $\frac{2}{3}$, the chance that a step leaves a given variable untouched.

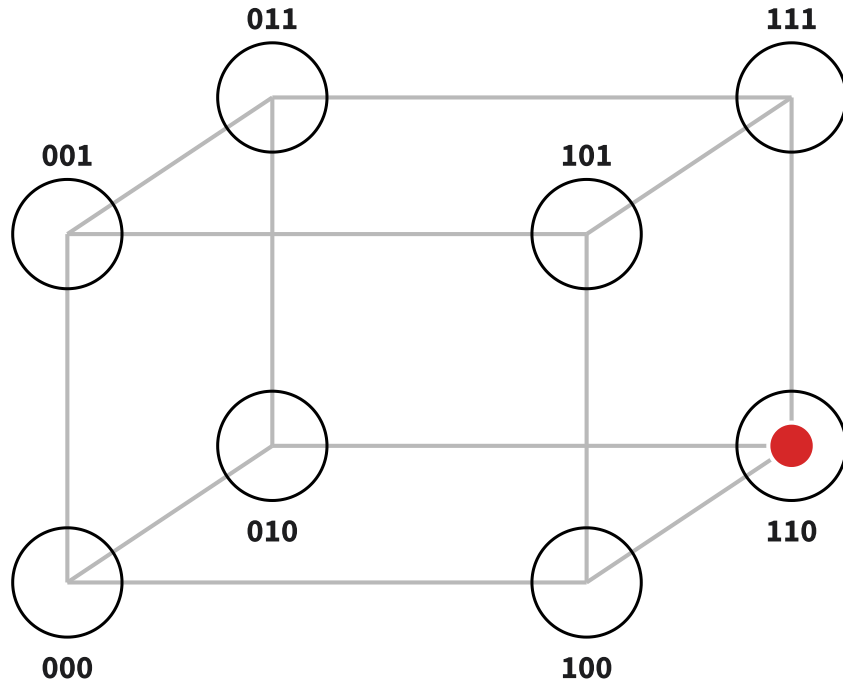
From any start the walkers' distribution converges, if the chain is irreducible and aperiodic



Theorem 6 (convergence). A finite chain that is **irreducible** (every state can reach every other) and **aperiodic** (not locked into returning only at multiples of some period $d \geq 2$) has exactly one stationary π , and $\mu_t \rightarrow \pi$ from every start.

- **Fair coin:** irreducible (flips connect the cube) and aperiodic (it can stay put). **Its distance falls to 0.**
- **Always flip:** the walkers alternate between the 4 states with an odd number of ones and the 4 with an even number, so half of them are always misplaced: distance ≥ 0.5 (**period 2**).

One walker followed for a long time is as good as many walkers



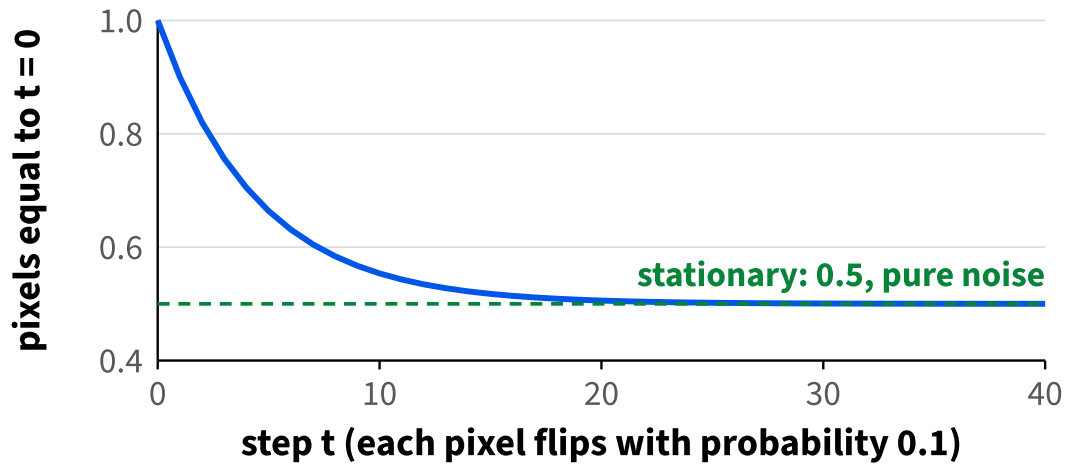
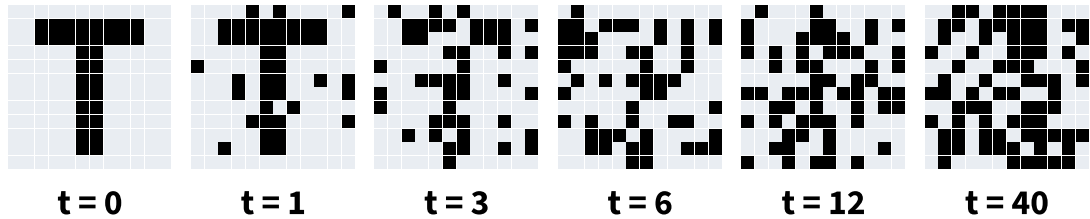
step **10000** · distance of the time fractions to uniform: **0.008**

Theorem 7 (ergodic theorem). For an irreducible finite chain with stationary π , from any start, with probability 1,

$$\frac{1}{N} \sum_{t=1}^N g(X_t) \longrightarrow \mathbb{E}_{\pi}[g] \quad \text{as } N \rightarrow \infty.$$

- With $g = 1[X = x]$: **the fraction of time at x tends to $\pi(x)$.** Left: one fair-coin walker, $\pi(x) = \frac{1}{8}$.
- One long run **replaces independent draws** from p (which need Z). **Price:** the steps are correlated, so N steps are worth fewer than N draws.
- **Plan:** build P with $\pi = p$, run one chain, average along the way.

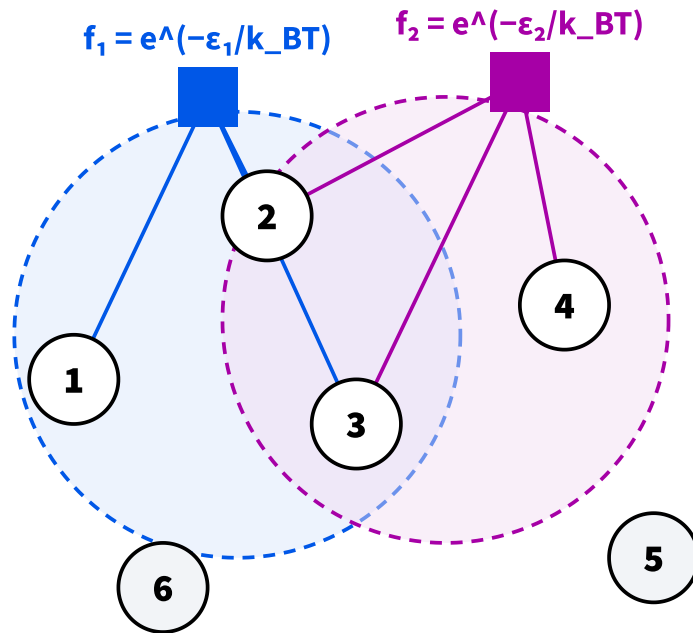
Diffusion models run a Markov chain into its stationary distribution, then back



- **Forward process:** a Markov chain that **adds noise**; here each pixel flips with probability 0.1 per step. On our cube: flip each x_i with probability 0.1.
- Its **stationary distribution is pure noise**: every pixel a fair coin, as the fair-coin walk ends at $\frac{1}{8}$ per state. Images use Gaussian noise, text [MASK] tokens, crystals random atoms (MatterGen).
- **Generation** starts from that noise and **runs a learned chain backwards**. You built the forward chain in Laboratory 2.

Ho, Jain and Abbeel, *NeurIPS* 2020; Austin et al., *NeurIPS* 2021; Zeni et al., *Nature* 2025. Full entries on the reference slides.

A learned energy for atoms is a Boltzmann model that only a sampler can use



f_1 reads atoms 1, 2, 3; f_2 reads 2, 3, 4: overlapping, not pairwise

- A **state** is the list of atomic positions $r = (r_1, \dots, r_n)$, $r_i \in \mathbb{R}^3$; element types z_i are fixed.
- A network reads each atom's neighbourhood $\mathcal{R}_i$ and outputs a **local energy**: $E_\theta(r) = \sum_i \epsilon_\theta(\mathcal{R}_i)$, a sum of factors, as on a factor graph.
- At temperature T : $p_\theta(r) = e^{-E_\theta(r)/(k_BT)} / Z_\theta$, and **Z_θ is an integral over $3n$ coordinates.**
- **Which configurations occur?** Simulation runs a **Markov chain whose stationary distribution is p_θ** . Each move reads only local energies, like our walker.

Each chain settles at its own π ; we need $\pi = p$

1

Maximum
entropy

2

Moment
matching

3

Monte Carlo

4

Markov chains

5

Detailed
balance

6

Metropolis–
Hastings

7

Gibbs sampling

8

Mixing

WHAT WE HAVE

- A walk **forgets its start** and settles at π with $\pi P = \pi$ (Theorem 6, convergence).
- **One long run** gives averages under π (Theorem 7, the ergodic theorem).

WHAT BLOCKS US

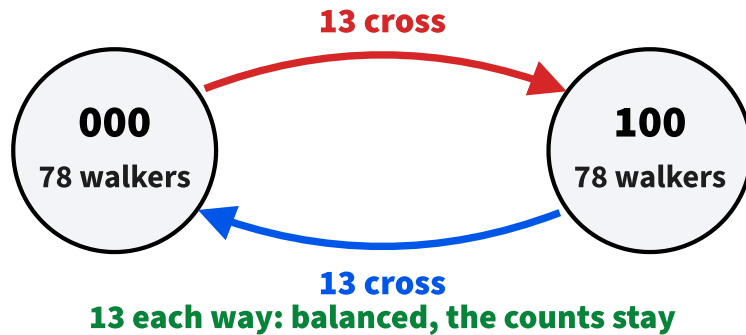
- The fair-coin walk settles at **uniform, not at p** .
- Checking $\pi P = \pi$ at one state sums over **all** states, and it needs $\pi = p$ itself, so **Z again**.

NEXT QUESTION Is there a check we can do one edge at a time, with ratios only?

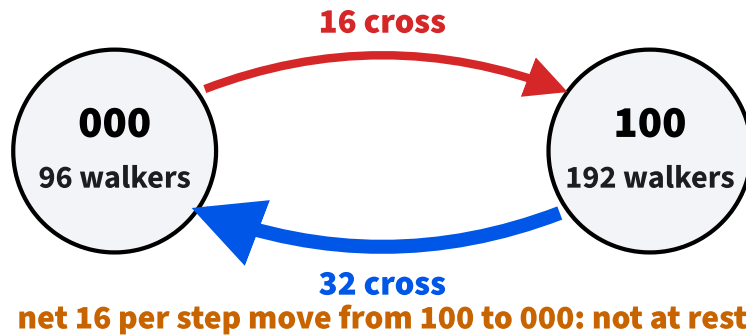
5 • Detailed balance

Count the walkers crossing one edge, and see why the fair-coin walk cannot keep p

Fair coin, walkers spread uniformly: 78 per state



Fair coin, walkers spread as p : 96 and 192



- **Setup: 624 walkers** (8×78 : whole counts). Count those crossing edge 000–100 in **one step**: each crosses with $\frac{1}{3}$ (it picks X_1) $\times \frac{1}{2}$ (the coin changes it) $= \frac{1}{6}$, either way.
- **Spread uniformly** (78 at each end): $78/6 = 13$ cross each way. **Balanced: the counts stay.**
- **Spread as p** ($624 \cdot \frac{12}{78} = 96$ at 000, 192 at 100): 16 cross one way, 32 back. **Net, 16 per step leave 100: the walkers cannot stay at p .**
- The share of walkers making a move is the **flow** $\pi(x)P(x, x')$. **Question:** to keep p , how often must a walk cross $000 \rightarrow 100$, compared with back?

Stationarity says: at every state, inflow equals outflow

Split $\pi P = \pi$ at a state x' into what arrives and what leaves:

$$\pi(x') = \sum_x \pi(x)P(x, x') \iff \underbrace{\sum_{x \neq x'} \pi(x)P(x, x')}_{\text{inflow into } x'} = \pi(x')(1 - P(x', x')) = \underbrace{\sum_{x'' \neq x'} \pi(x')P(x', x'')}_{\text{outflow from } x'}.$$

- This is **global balance**: one equation per state, and each equation sums over all states.
- For 2^{1024} states we can **neither write these equations down nor check them**.
- **Wanted**: a stronger condition that can be **checked edge by edge**.

Balance every edge, and every state is balanced automatically

Definition (detailed balance). A transition matrix P satisfies detailed balance with π if **the two flows on every edge are equal**:

$$\pi(x) P(x, x') = \pi(x') P(x', x) \quad \text{for all } x, x'.$$

Lemma 8. Detailed balance implies $\pi P = \pi$.

Proof. Sum the definition over x :

$$\sum_x \pi(x) P(x, x') = \sum_x \pi(x') P(x', x) = \pi(x') \sum_x P(x', x) = \pi(x') \cdot 1. \quad \blacksquare$$

Detailed balance needs only ratios, so Z cancels

Divide the definition, with $\pi = p = g/Z$:

$$\frac{P(x, x')}{P(x', x)} = \frac{p(x')}{p(x)} = \frac{g(x')/Z}{g(x)/Z} = \frac{g(x')}{g(x)}.$$

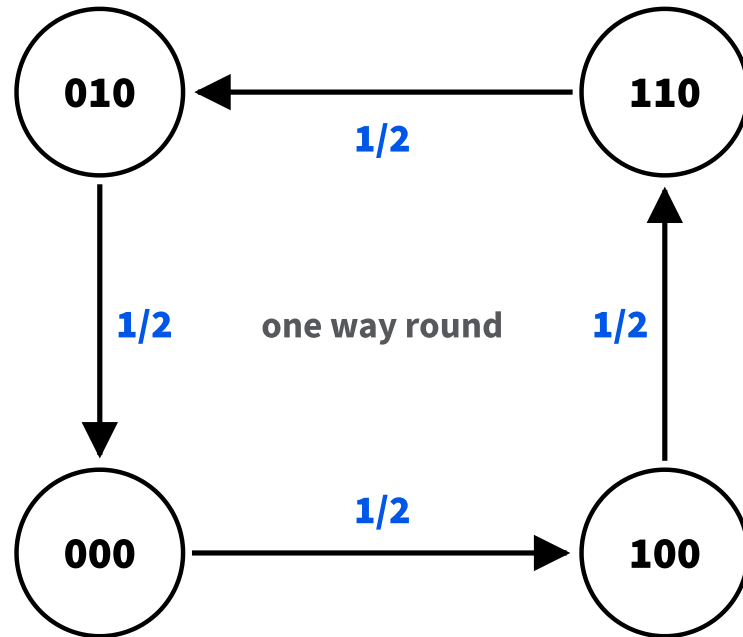
- **Answer for edge 000–100:** cross from 000 to 100 **twice as often** as back, since $\frac{g(100)}{g(000)} = \frac{f_{12}(1,0)f_{23}(0,0)}{f_{12}(0,0)f_{23}(0,0)} = \frac{8 \cdot 3}{4 \cdot 3} = 2$. Then 96 and 192 walkers send equal numbers.
- The fair-coin walk has $\frac{1}{6}$ both ways, **ratio 1: it fails**, which is why its walkers went to uniform.
- Neighbours differ in one variable, so the ratio reads only the factors touching it: above, $f_{23}(0, 0)$ cancels and $\frac{f_{12}(1,0)}{f_{12}(0,0)} = \frac{8}{4}$ is left. **Local and cheap.**

Edge by edge on our chain: the ratio each edge must have

edge $x-x'$	$g(x)$	$g(x')$	needed $\frac{P(x,x')}{P(x',x)}$	fair coin
000-100	12	24	2	1
000-001	12	4	1/3	1
000-010	12	1	1/12	1
001-011	4	4	1	1
001-101	4	8	2	1
010-011	1	4	4	1
010-110	1	5	5	1
011-111	4	20	5	1
100-101	24	8	1/3	1
100-110	24	5	5/24	1
101-111	8	20	5/2	1
110-111	5	20	4	1

- Detailed balance with p fixed, on each edge, **only the ratio** of the two directions: $g(x')/g(x)$.
- The probabilities themselves are **free**: on 000-100, $\frac{1}{3}$ there and $\frac{1}{6}$ back works; so do $\frac{1}{6}$ and $\frac{1}{12}$, only slower.
- The fair-coin walk has ratio 1 everywhere: it **meets only edge 001-011**.
- Twelve ratios from the weights; **none needs $\mathbb{Z}$** .
Next: one recipe that turns them into actual probabilities.

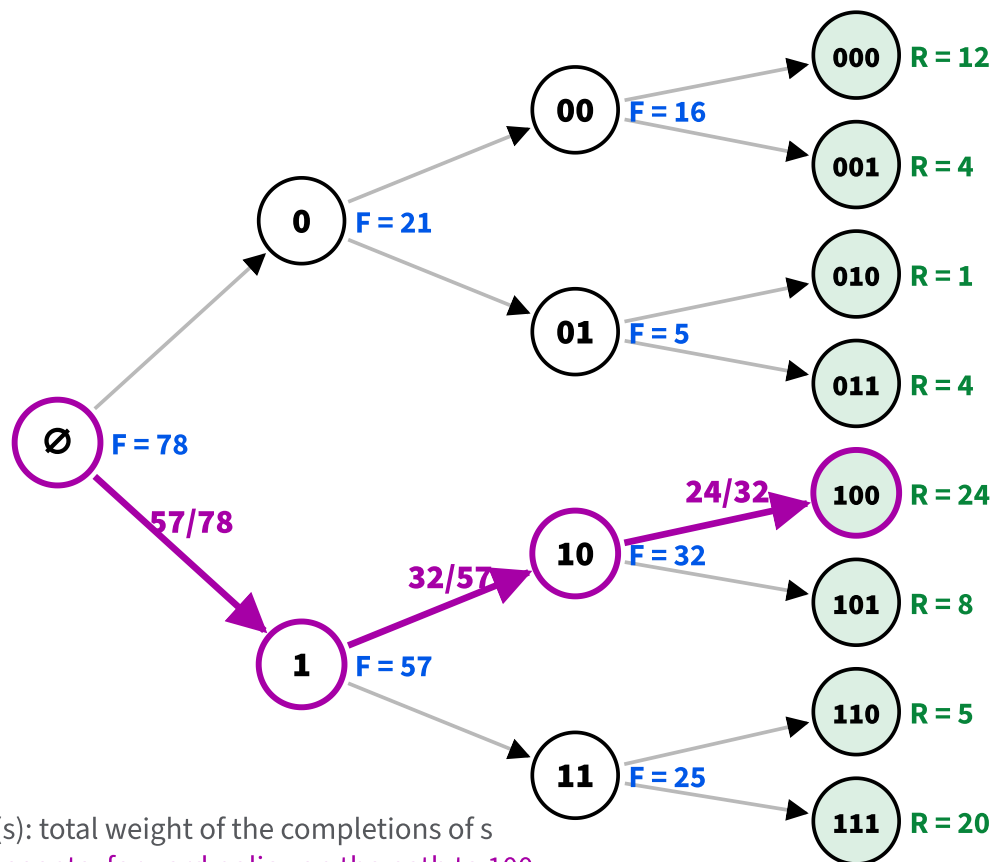
Detailed balance is sufficient, not necessary



- A walk on one face of the cube (4 states): each step, stay with probability $\frac{1}{2}$, or **move on one way round**, $000 \rightarrow 100 \rightarrow 110 \rightarrow 010 \rightarrow 000$.
- Spread the walkers uniformly over the **four** states: $\frac{1}{4}$ **at each**. Each state sends half of its $\frac{1}{4}$, that is $\frac{1}{8}$ of all walkers, and receives $\frac{1}{8}$ from the state before it: **uniform is stationary**.
- **But no edge balances**: $000 \rightarrow 100$ carries $\frac{1}{8}$, $100 \rightarrow 000$ carries 0.
- So stationarity does **not** need detailed balance. We use detailed balance because it is enough, and it can be **checked edge by edge**.

A chain with detailed balance is also called **reversible**: on every edge, as many jumps one way as the other.

GFlowNets build an object step by step, choosing each step by flow



$F(s)$: total weight of the completions of s
 magenta: forward policy on the path to 100

- **Goal:** sample an object x (a molecule, a reasoning chain) with probability $R(x)/Z$, Z unknown. Our chain: $R = g$; choose x_1 , then x_2 , then x_3 (left).
- **Flow** $F(s)$: the total weight of all completions of a partial choice s . $F(1) = 24 + 8 + 5 + 20 = 57$, $F(10) = 24 + 8 = 32$, $F(\emptyset) = 78$.
- **Choose each next value in proportion to its flow:** $P_F(s' | s) = F(s')/F(s)$.
- Path to 100 (magenta): $\frac{57}{78} \cdot \frac{32}{57} \cdot \frac{24}{32} = \frac{24}{78}$. The fractions cancel: **every x comes out with probability $g(x)/78$.**

Y. Bengio et al. GFlowNet foundations. *JMLR* 24(210), 2023.

GFlowNets turn a balance condition into a training loss

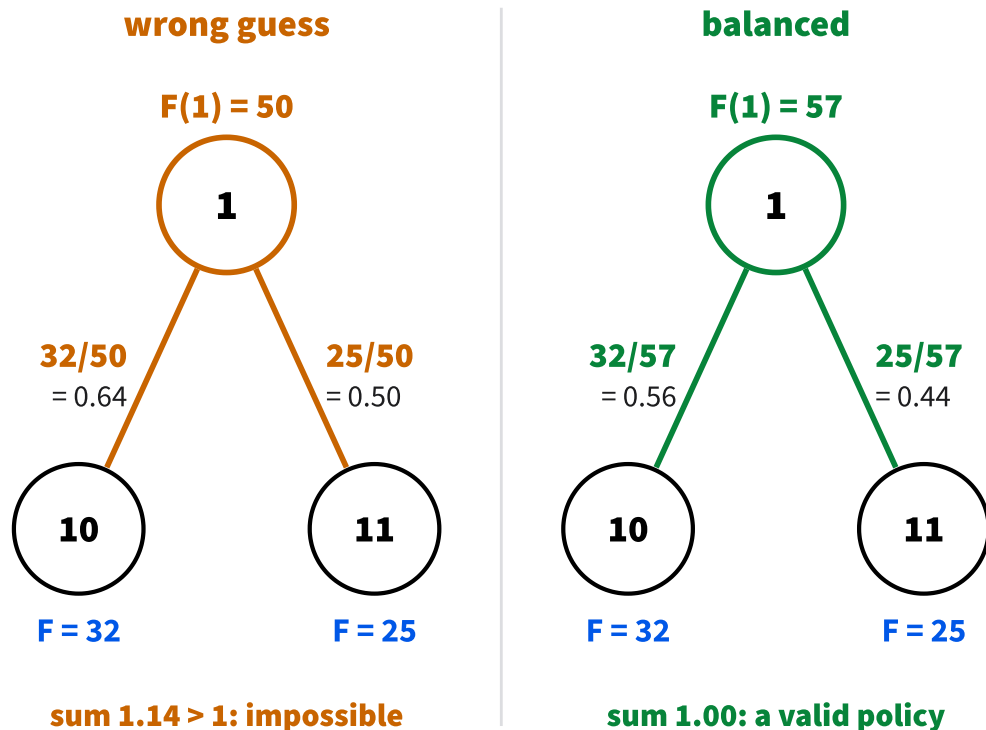
The rule $P_F(s' | s) = F(s')/F(s)$ is **balance on each edge** of the tree:

$$F(s) P_F(s' | s) = F(s'),$$

the shape of detailed balance, $\pi(x)P(x, x') = \pi(x')P(x', x)$.

- **Large problems:** the tree is too big to sum, so F and P_F are **networks**, trained on sampled edges (with $F = R$ at complete objects) to shrink $(\ln F(s) + \ln P_F(s' | s) - \ln F(s'))^2$.
- **Why it works** (left): a wrong guess $F(1) = 50$ needs $P_F = \frac{32}{50}$ and $\frac{25}{50}$ on the two edges, which sum to 1.14. No policy fits, so **the loss stays above 0 until $F(1) = 32 + 25 = 57$** .

For language models: Hu et al., *ICLR* 2024.



We know what every edge must satisfy; now we need a recipe that satisfies it

1
Maximum
entropy

2
Moment
matching

3
Monte Carlo

4
Markov chains

5
Detailed
balance

6
Metropolis–
Hastings

7
Gibbs sampling

8
Mixing

WHAT WE HAVE

- A target for every edge: $P(x, x')/P(x', x) = g(x')/g(x)$.
- Meeting it on every edge makes p stationary (Lemma 8).

WHAT IS MISSING

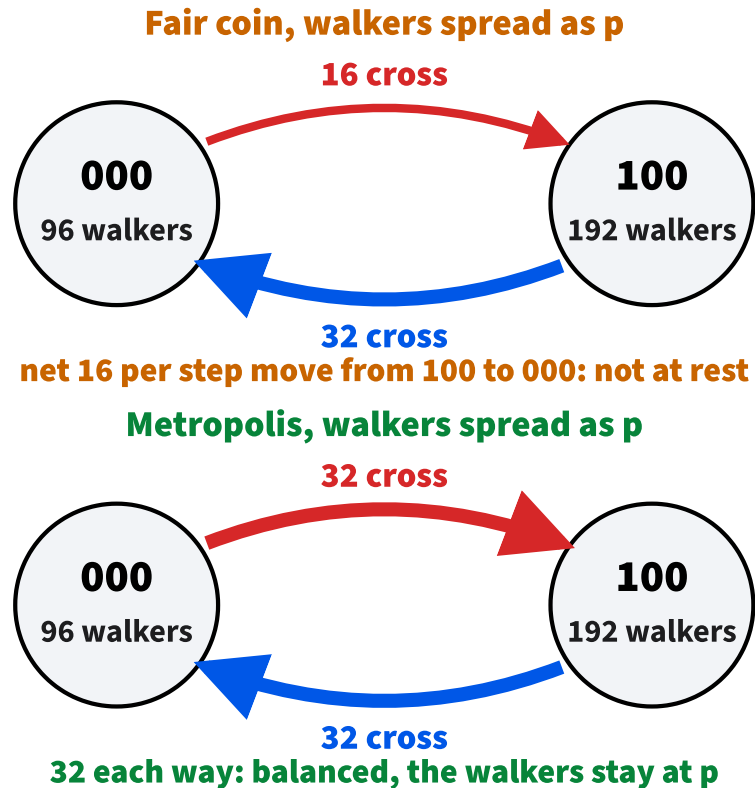
- An actual walk with these ratios: **000 → 100 twice as often as 100 → 000**, and the required ratio on each of the cube's 12 edges.
- A recipe that builds such a walk for **any** weights g .

NEXT QUESTION Propose a move as the fair coin does, then accept it with just the right probability? **6 • Metropolis–Hastings**

Ten minutes

After the break: an algorithm from 1953 that still runs inside physics simulations and Bayesian statistics, and that a 2025 study used to make a language model reason better without any training.

Propose like the fair coin, then accept just often enough to balance the edge



- **One step of the new walk: propose**, that is, pick one of the three variables (probability $\frac{1}{3}$) and plan to flip it; then **accept** the flip, or stay.
- Edge 000–100, 624 walkers spread as p . A walker proposes this edge when it picks X_1 :

	walkers	propose ($\frac{1}{3}$)	accept	cross
000 → 100	96	32	all	32
100 → 000	192	64	half	32

- **Equal: balanced.** Rule: accept a move to a heavier state always; to a lighter one with probability $g(\text{lighter})/g(\text{heavier})$, here $12/24 = \frac{1}{2}$:

$$\alpha(x, x') = \min \left(1, \frac{g(x')}{g(x)} \right) \quad (\text{Metropolis et al., 1953}).$$

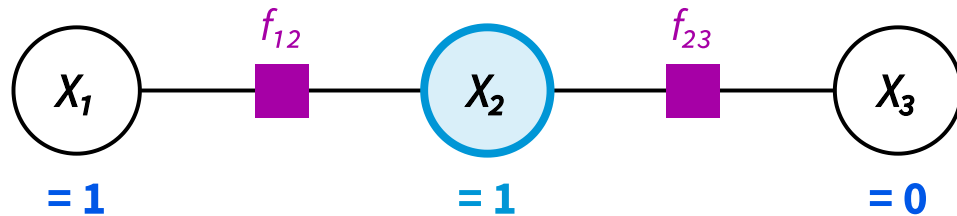
The rule balances every edge, not only 000–100

Each step: $P(x, x') = \frac{1}{3} \alpha(x, x')$ for each neighbour x' . Take any edge and call x' its **heavier** end, $g(x') \geq g(x)$:

$$\begin{aligned} \text{flow } x \rightarrow x' : \quad & \frac{g(x)}{Z} \cdot \frac{1}{3} \cdot \underbrace{1}_{\alpha(x, x')} = \frac{g(x)}{3Z}, \\ \text{flow } x' \rightarrow x : \quad & \frac{g(x')}{Z} \cdot \frac{1}{3} \cdot \underbrace{\frac{g(x)}{g(x')}}_{\alpha(x', x)} = \frac{g(x)}{3Z}. \end{aligned}$$

- **Equal on every edge**, so p is stationary (Lemma 8: balanced edges give $\pi P = \pi$). On edge 000–100: $12/234 = \frac{4}{78}$ each way.
- The walker only ever computes $g(x')/g(x)$: **Z is never used**.

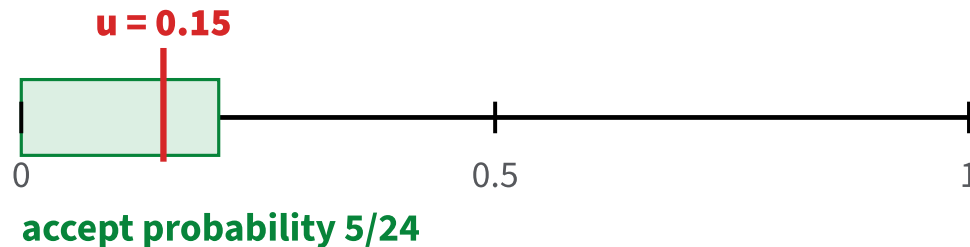
Three proposals on the chain: accepted, rejected, and a move to a lighter state accepted



at 100, propose: flip $X_2 \rightarrow 110$, weight 5

$$\text{ratio } f_{12}(1,1)f_{23}(1,0) / f_{12}(1,0)f_{23}(0,0) = 5/24 = 0.21$$

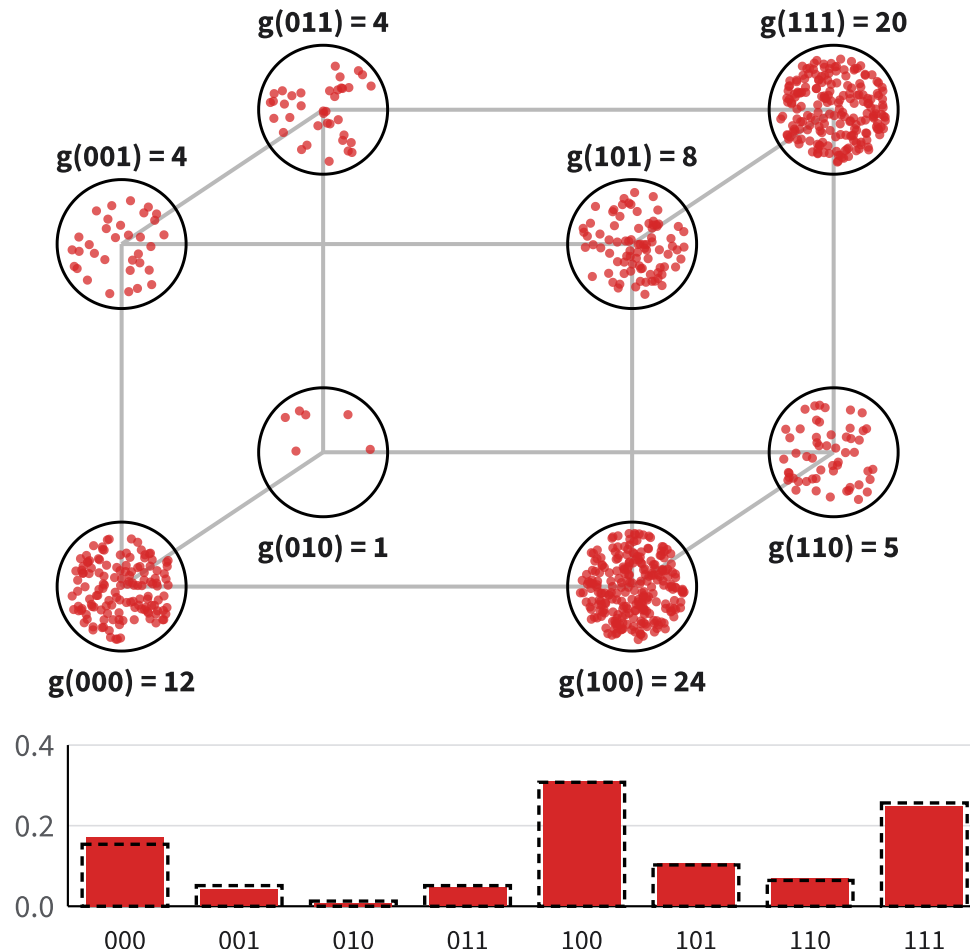
u : a fresh random number, uniform on $[0, 1]$; accept if it lands in the green part



$u = 0.15 < 0.21$: accept, move to 110

- **Propose:** flip one variable at random.
- **Ratio** $g(x')/g(x)$: only the flipped variable's factors; no Z .
- **Accept** if u falls below $\min(1, \text{ratio})$, where u is a fresh random number, uniform on $[0, 1]$, for each proposal (here 0.47, 0.62, 0.15).
- **Rejected:** stay, and count the current state again.
- **Lighter states are sometimes accepted:** that is how a walker leaves a peak.

A new acceptance rule, and the walkers settle at p



- Same 800 walkers, same start 010. **Only one change** from the fair-coin walk: accept with $\min(1, g(x')/g(x))$.
- **Distance to p :** 0.987, 0.731, 0.222, 0.118, 0.052, 0.028, ... $\rightarrow 0$.
- The red bars **fill the dashed outlines** $p = (12, 4, 1, 4, 24, 8, 5, 20)/78$.
- **The walkers never used $Z = 78$.**

step **30** · distance of the walkers to p : **0.029**

With an uneven proposal, Hastings adds the ratio of proposal probabilities

Definition (Metropolis–Hastings). Target $\pi \propto g$, proposal $q(x' | x)$. From the current state x :

1. propose $x' \sim q(\cdot | x)$;
 2. accept with probability $\alpha(x, x') = \min \left\{ 1, \frac{g(x') q(x | x')}{g(x) q(x' | x)} \right\}$;
 3. otherwise stay at x , and count x again.
- A symmetric proposal, $q(x' | x) = q(x | x')$, **gives back the rule of 1953.**
 - The target enters only through $g(x')/g(x)$: **Z is never needed.**

N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, E. Teller. Equation of state calculations by fast computing machines. *J. Chem. Phys.* 21(6), 1087–1092, 1953. W. K. Hastings. Monte Carlo sampling methods using Markov chains and their applications. *Biometrika* 57(1), 97–109, 1970.

Metropolis–Hastings satisfies detailed balance for every target and every proposal

Theorem 9. The Metropolis–Hastings transition matrix satisfies detailed balance with π . So π is stationary (Lemma 8: balanced edges give $\pi P = \pi$).

Proof. For $x \neq x'$, move $\pi(x)q(x' | x)$ inside the minimum:

$$\pi(x)P(x, x') = \pi(x)q(x' | x) \min \left\{ 1, \frac{\pi(x')q(x | x')}{\pi(x)q(x' | x)} \right\} = \min \{ \pi(x)q(x' | x), \pi(x')q(x | x') \}.$$

The right side does not change when x and x' swap, so it also equals $\pi(x')P(x', x)$. ■

The whole transition matrix of our chain, in exact fractions

from \ to	000	001	010	011	100	101	110	111
000	19/36	1/9	1/36	0	1/3	0	0	0
001	1/3	0	0	1/3	0	1/3	0	0
010	1/3	0	0	1/3	0	0	1/3	0
011	0	1/3	1/12	1/4	0	0	0	1/3
100	1/6	0	0	0	47/72	1/9	5/72	0
101	0	1/6	0	0	1/3	1/6	0	1/3
110	0	0	1/15	0	1/3	0	4/15	1/3
111	0	0	0	1/15	0	2/15	1/12	43/60
weight g	12	4	1	4	24	8	5	20

- **How to read it:** row = from, column = to. Bold: the 3 neighbours; grey: stay; 0: not a neighbour. Bottom row: the weights g .
- **Row 000:** $P(000, x') = \frac{1}{3} \min(1, \frac{g(x')}{12})$. To 100: $\frac{1}{3} \min(1, \frac{24}{12}) = \frac{1}{3}$; to 001: $\frac{1}{3} \cdot \frac{4}{12} = \frac{1}{9}$; to 010: $\frac{1}{3} \cdot \frac{1}{12} = \frac{1}{36}$. Stay: $1 - \frac{1}{3} - \frac{1}{9} - \frac{1}{36} = \frac{19}{36}$.
- **Column 000 against the weights:** $12 \cdot \frac{19}{36} + 4 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 24 \cdot \frac{1}{6} = 12 = g(000)$. Every column returns its weight: $pP = p$.

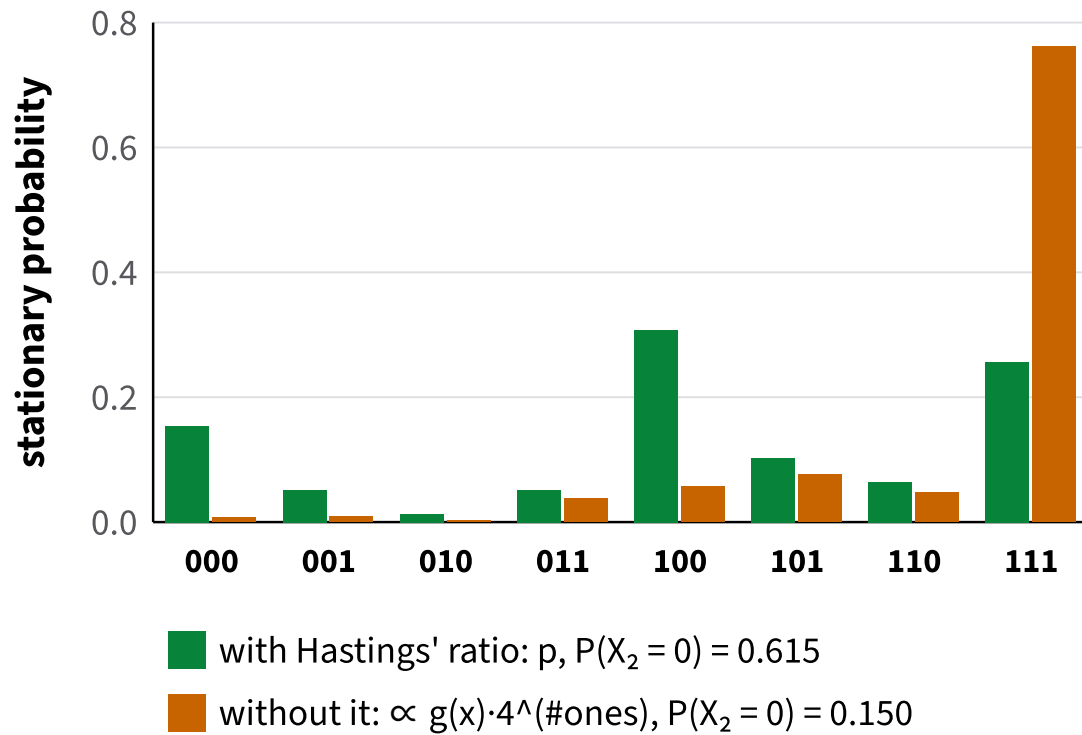
Downhill always; uphill with probability $e^{-\Delta E/T}$

Recall: energy $E(x) = -\ln g(x)$, and at temperature T the Boltzmann distribution $p_T(x) = e^{-E(x)/T} / Z_T$. **A heavy state has low energy.**

$$\frac{p_T(x')}{p_T(x)} = e^{-(E(x')-E(x))/T} \implies \alpha = \min(1, e^{-\Delta E/T}), \quad \Delta E = E(x') - E(x).$$

- **From 100 to 101:** $\Delta E = -\ln 8 + \ln 24 = \ln 3$. At $T = 1$: accept with $e^{-\ln 3} = \frac{1}{3}$, the rejected proposal of the step-by-step slide.
- At $T = 0.2$: $e^{-5 \ln 3} = \frac{1}{243} = 0.0041$. **Cold chains almost never climb:** a walker at a peak stays there for a long time.

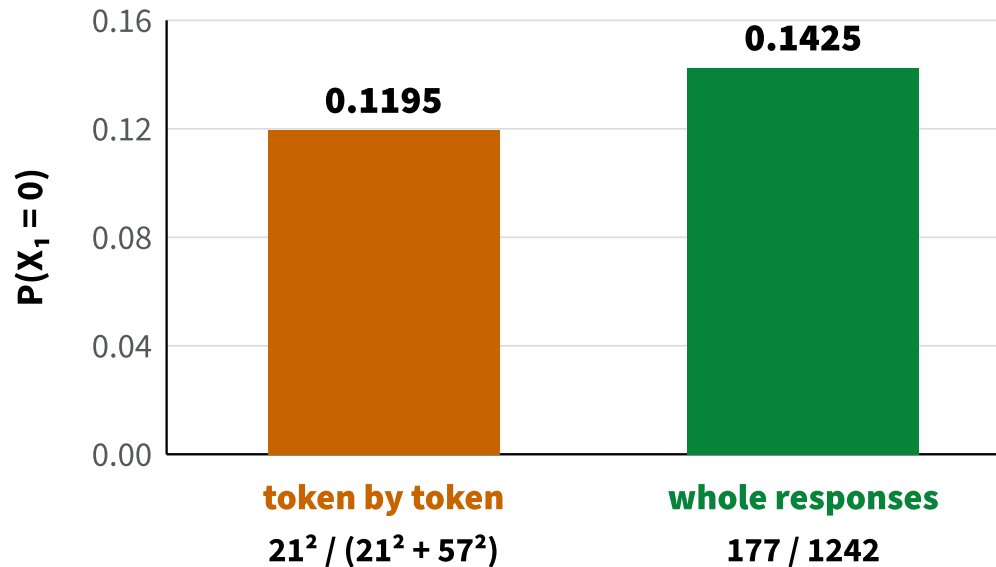
Drop Hastings' correction and the chain converges, confidently, to the wrong answer



- **Uneven proposal:** pick a variable, propose the value 1 with probability 0.8 and 0 with 0.2.
- **With** the ratio $q(x | x')/q(x' | x)$ in α : the chain settles at $p, P(X_2 = 0) = 0.615$.
- **Without it:** each $0 \rightarrow 1$ proposal is $0.8/0.2 = 4$ times as likely as its reverse, and nothing corrects this. The walk balances $g(x) \cdot 4^{\#ones(x)}$ instead: $P(X_2 = 0) = 0.150$.
- It converges smoothly and looks healthy. Running longer does not help: **the target is wrong.**

Sharpen a language model: raise whole responses to a power, not single tokens

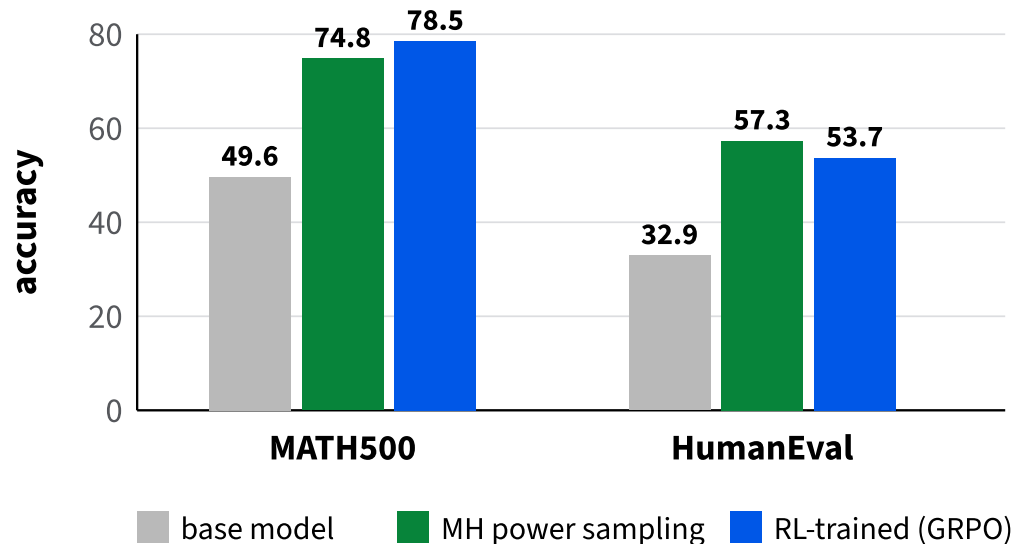
Our chain as a 3-token response, $\alpha = 2$



- **Goal:** sample responses y from $p(y)^\alpha / Z_\alpha$, $\alpha > 1$: responses the model already finds likely become likelier. $Z_\alpha = \sum_y p(y)^\alpha$ is out of reach.
- **Our chain as a 3-token response**, $\alpha = 2$. The first token is 0 with probability $\frac{12+4+1+4}{78} = \frac{21}{78}$, and 1 with $\frac{57}{78}$.
- **Token by token** (square each next-token probability, renormalize): $\frac{21^2}{21^2+57^2} = 0.1195$.
- **Whole responses** (square each $g(x)$, renormalize): $\frac{12^2+4^2+1^2+4^2}{\sum_x g(x)^2} = \frac{177}{1242} = 0.1425$. **Squaring a sum is not summing squares**, so token-by-token sharpening samples a different distribution.

Metropolis–Hastings samples the sharpened model without Z_α , and it reasons better

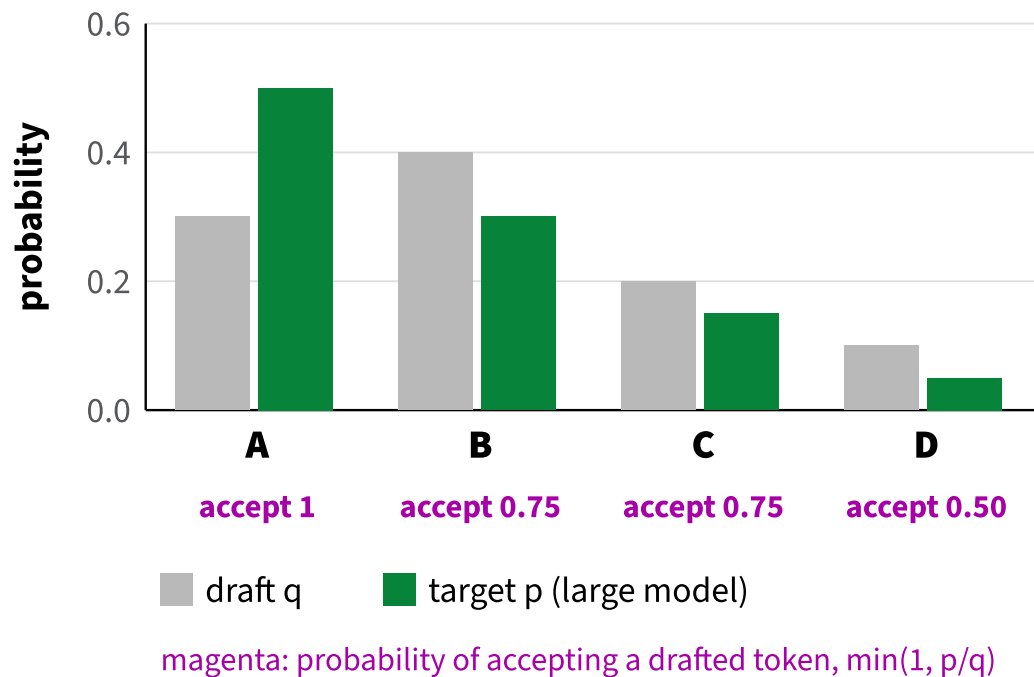
Karan & Du (2025), Qwen2.5-Math-7B: accuracy (%)



- **Why Metropolis–Hastings:** it needs only the ratio $p(y')^\alpha / p(y)^\alpha$, and the base model gives $p(y)$ of any response as the product of its token probabilities. Z_α **cancels.**
- **Propose:** keep a random prefix of the current response and regenerate the rest with the base model. **Accept** with $\min\{1, \frac{p(y')^\alpha q(y|y')}{p(y)^\alpha q(y'|y)}\}$, Hastings' correction for this uneven proposal.
- **Result** (left, $\alpha = 4$): **no training**, and accuracy close to reinforcement learning; MATH500: 49.6 $\rightarrow$ 74.8, against 78.5 for RL.

A. Karan, Y. Du. Reasoning with sampling: your base model is smarter than you think. arXiv:2510.14901, 2025.

Speculative decoding: a small model drafts, the large model accepts or rejects



- **Problem:** a large model p produces one token per slow, sequential pass. A small model q is fast but worse.
- **Idea:** the small model **proposes** tokens; the large model **checks** several of them in one pass and **accepts** each with $\min(1, p(x)/q(x))$: the Metropolis–Hastings acceptance for target p and proposal q .
- After a rejection, a replacement is drawn from the **leftover** $\max(0, p - q)$, normalized. Next slide: why.
- **Benefit:** most tokens come from the fast model, yet the output is **exactly** the large model's p : 2 to 3 times faster, same quality.

Y. Leviathan, M. Kalman, Y. Matias. Fast inference from transformers via speculative decoding. *ICML* 2023.

Why speculative decoding outputs exactly the large model's distribution

token	q	p	accept	$\min(p, q)$	leftover
A	0.30	0.50	1	0.30	0.20
B	0.40	0.30	0.75	0.30	0
C	0.20	0.15	0.75	0.15	0
D	0.10	0.05	0.50	0.05	0
total	1	1		0.80	0.20

$\text{accept} = \min(1, p/q)$; $\text{leftover} = \max(0, p - q)$.

- **Accepted part:** token x is drafted and kept with probability $q(x) \min(1, \frac{p(x)}{q(x)}) = \min(p(x), q(x))$. Total 0.80, so a rejection has probability 0.20.
- **Missing part:** $p - \min(p, q)$, the leftover. It also totals 0.20, and here all of it is on A.
- **After a rejection, draw from the leftover:** A with probability 1. Output: A $0.30 + 0.20 = 0.50$, B 0.30, C 0.15, D 0.05: **exactly** p .

What Metropolis–Hastings does not solve

1. Rejections waste work. At 000 the walker stays $\frac{19}{36}$ of the time. Small moves are accepted but explore slowly; large moves explore fast but are mostly rejected.

2. Steps are correlated. Each step changes one variable, so N steps are worth fewer than N independent draws.

3. Barriers. An uphill move is accepted with $e^{-\Delta E/T}$, only $\frac{1}{243}$ at $T = 0.2$: a cold walker stays in one peak for a long time.

4. No stopping rule. Convergence is guaranteed only as the number of steps grows; the chain never signals that it has arrived.

Next, Gibbs sampling removes the rejections. The end of today measures correlation, barriers and when to stop.

On a factor graph a one-variable move reads only that variable's factors

1
Maximum
entropy

2
Moment
matching

3
Monte Carlo

4
Markov chains

5
Detailed
balance

6
Metropolis–
Hastings

7
Gibbs sampling

8
Mixing

WHAT WE HAVE

- A valid sampler for any target, from ratios only.
- Flipping X_2 reads only f_{12} and f_{23} .

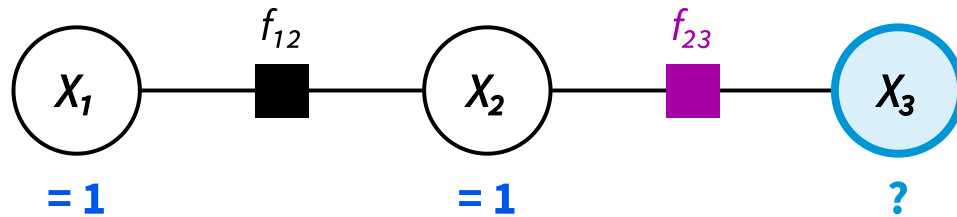
WHAT IS WASTEFUL

- Rejections: at 000 the walker stays $\frac{19}{36}$ of the time.
- We compute the weights of both values of the variable, then throw a proposal away.

NEXT QUESTION Why not draw the variable's new value directly from its exact conditional?

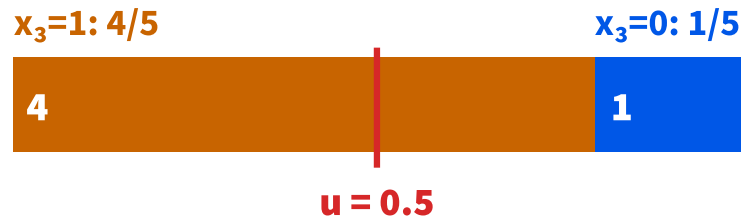
7 • Gibbs sampling

Redraw one variable from its conditional: only its factors are read



redraw X_3 given $X_2 = 1$: $f_{23}(1,0) = 1$, $f_{23}(1,1) = 4$

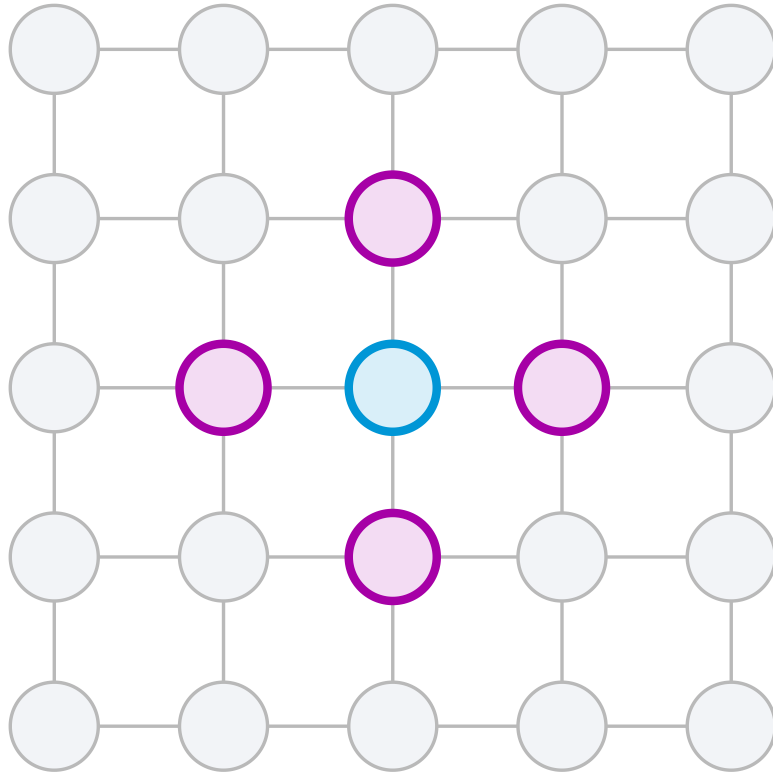
$u = 0.5 < 4/5$: set $X_3 = 1$, state 111



two updates: from peak 100 to peak 111

- **Pick a variable**; hold all the others fixed.
- Its two possible values have weights read from **the factors that touch it**, here 24 and 5.
- Normalize over **two values**, not over 2^n states: $24 + 5 = 29$.
- Draw the new value with those probabilities. **Nothing is ever rejected.**
- **Two updates** carried the walker from peak 100 to peak 111.

Every factor that does not touch the variable cancels from its conditional



blanket of the centre: its 4 neighbours

$$\begin{aligned}
 p(x_2 \mid x_1, x_3) &= \frac{g(x_1, x_2, x_3)}{\sum_{x'_2} g(x_1, x'_2, x_3)} \\
 &= \frac{f_{12}(x_1, x_2) f_{23}(x_2, x_3)}{\sum_{x'_2} f_{12}(x_1, x'_2) f_{23}(x'_2, x_3)}.
 \end{aligned}$$

- In general $g(x) = \prod_a f_a$. A factor that does not contain x_i is the same number above and below, **so it cancels**.
- What remains are the factors touching x_i ; their other variables form the **Markov blanket** of x_i .

On a grid: **four neighbours, however large the grid.**

Gibbs sampling is Metropolis–Hastings that always accepts

Definition (Gibbs sampler). Repeat: choose a variable i (at random, or in turn); draw a new value $x_i \sim p(x_i \mid x_{-i})$, which **reads only the factors containing x_i** .

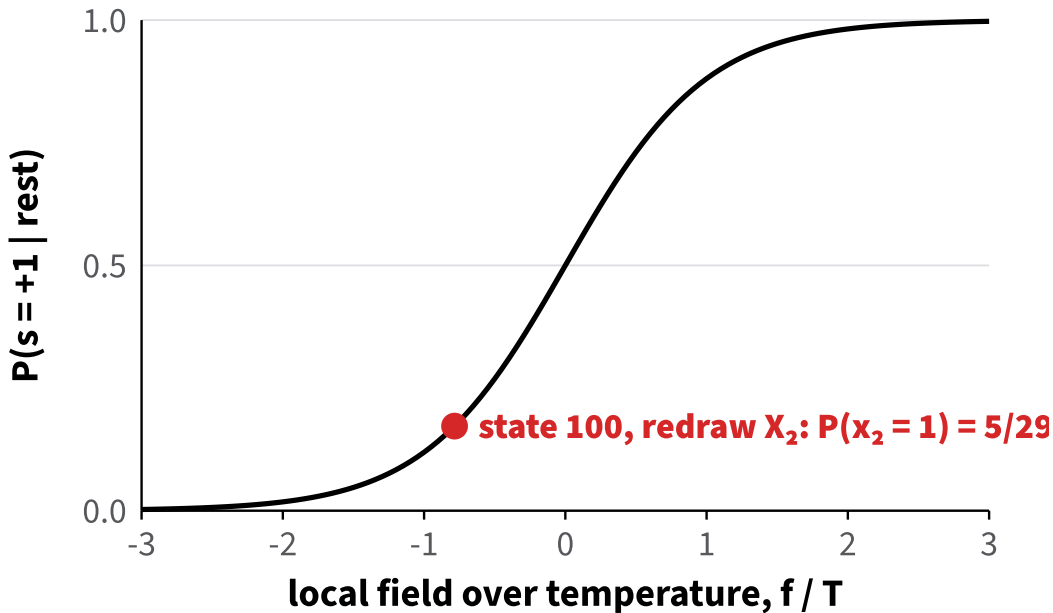
Theorem 10. Proposing $x' = (x'_i, x_{-i})$ with $q(x' \mid x) = p(x'_i \mid x_{-i})$ gives Metropolis–Hastings **acceptance 1**, so p is stationary.

Proof. Only x_i changes. Split each joint as (the others) $\times$ (the conditional of x_i); top and bottom then hold **the same three factors**:

$$\frac{p(x') q(x \mid x')}{p(x) q(x' \mid x)} = \frac{\overbrace{p(x_{-i}) p(x'_i \mid x_{-i})}^{p(x')} \cdot \overbrace{p(x_i \mid x_{-i})}^{q(x \mid x')}}{\underbrace{p(x_{-i}) p(x_i \mid x_{-i})}_{p(x)} \cdot \underbrace{p(x'_i \mid x_{-i})}_{q(x' \mid x)}} = 1. \quad \blacksquare$$

Our chain, from 100 redraw X_2 to get 110: $\frac{5}{24} \cdot \frac{24/29}{5/29} = 1$. Geman and Geman, *IEEE TPAMI*, 1984.

For spins, the conditional is a logistic function of the local field



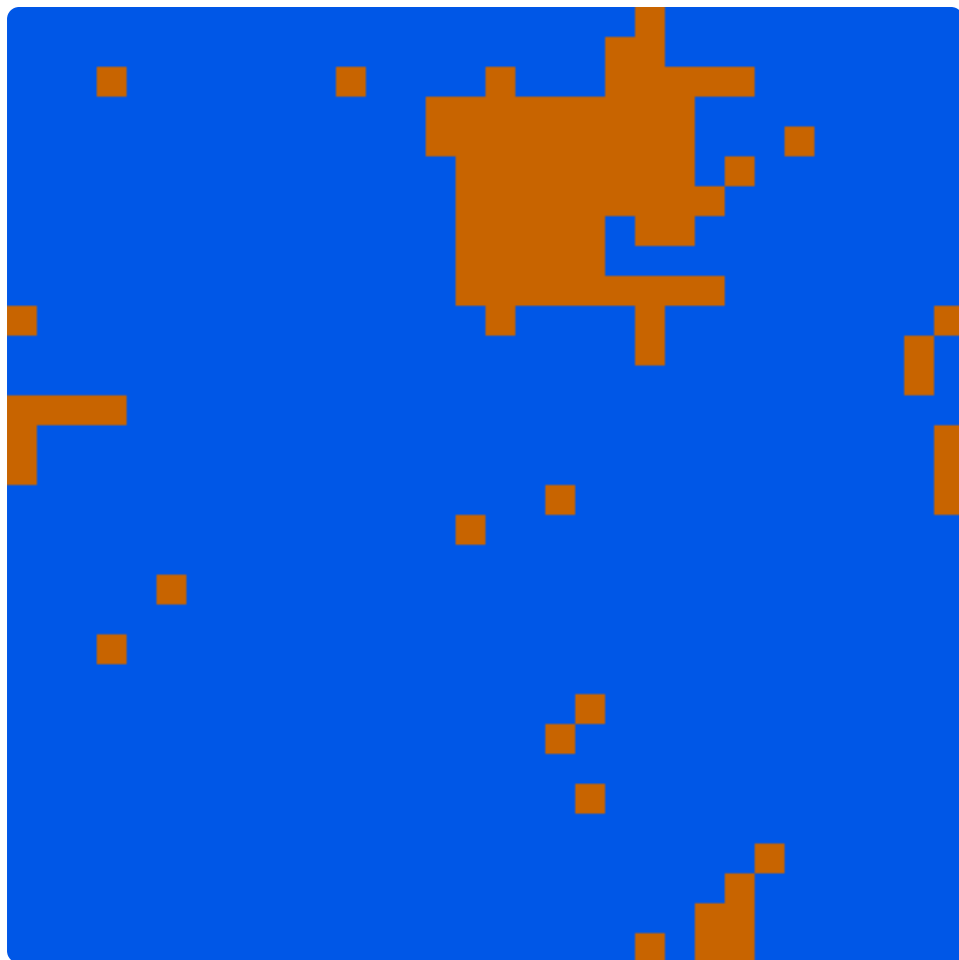
Spins $s_i = \pm 1$, energy $E(s) = -\sum_{i < j} J_{ij} s_i s_j - \sum_i h_i s_i$, $p_T \propto e^{-E/T}$. The terms containing s_i are $-f_i s_i$, with **local field** $f_i = \sum_j J_{ij} s_j + h_i$: what the neighbours say.

$$\frac{P(s_i = +1 \mid \text{rest})}{P(s_i = -1 \mid \text{rest})} = \frac{e^{f_i/T}}{e^{-f_i/T}},$$

$$P(s_i = +1 \mid \text{rest}) = \frac{1}{1 + e^{-2f_i/T}}.$$

Check on the chain. The walker of the first Gibbs slide sits at state 100 ($x_1 = 1, x_2 = 0, x_3 = 0$) and redraws X_2 . The weights 24 ($x_2 = 0$) and 5 ($x_2 = 1$) give $P(x_2 = 1) = \frac{5}{29}$: the red dot on the curve.

A 32×32 grid: each update reads four neighbours

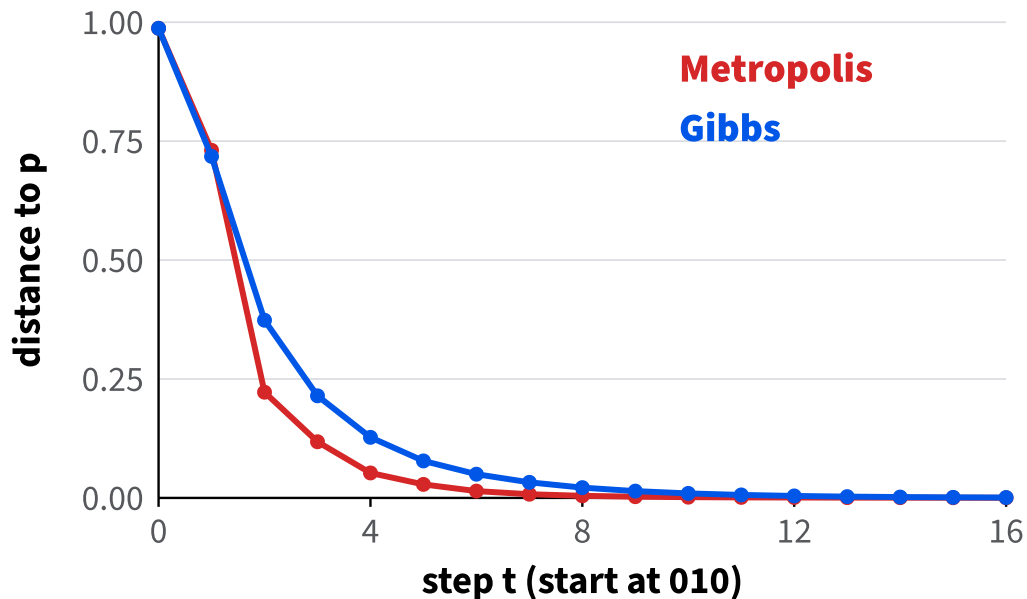


$T = 2.00$ · sweeps **60** · mean spin **0.807**

- 2^{1024} **states**. Energy $E(s) = -\sum_{\text{neighbours}} s_i s_j$: neighbours prefer equal spins. Blue = $+1$, orange = -1 .
- Update each spin by the logistic rule, with local field $f_i = \text{the sum of its 4 neighbouring spins}$ ($J = 1, h = 0$): from -4 to $+4$. This is **Glauber dynamics**.
- **One sweep** = 1,024 updates, each 4 additions and one random number.
- **Critical temperature** $T_c = 2 / \ln(1 + \sqrt{2}) = 2.269$. **Below** T_c : large patches of equal spins grow, and the mean spin heads to $+1$ or -1 : two modes, like our two peaks. **Above** T_c : the spins stay mixed, mean spin near 0.

R. J. Glauber. *J. Math. Phys.* 4(2), 294–307, 1963. $T_c = 2 / \ln(1 + \sqrt{2})$; L. Onsager. *Phys. Rev.* 65, 117–149, 1944.

On our chain, Metropolis reaches p faster than Gibbs



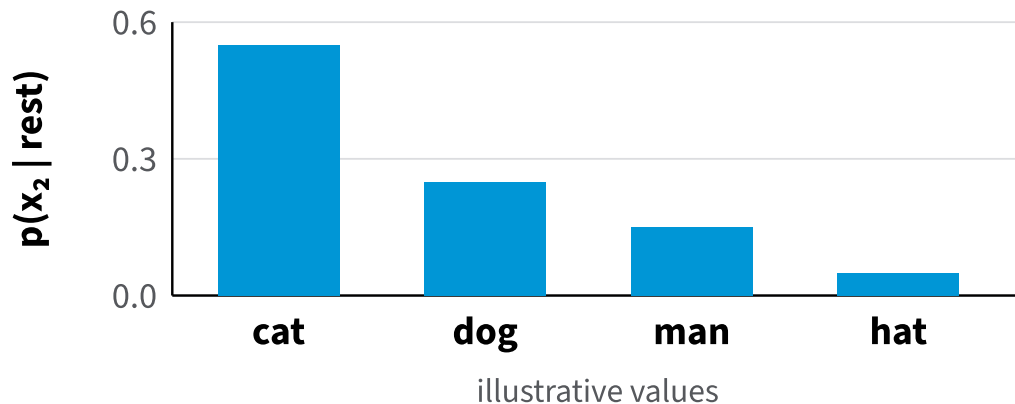
- **Random-scan Gibbs:** pick a variable ($\frac{1}{3}$), redraw it from its two weights. From 000, pick X_1 : 12 against 24, so to 100 with $\frac{1}{3} \cdot \frac{24}{36} = \frac{2}{9}$ (Metropolis: $\frac{1}{3}$). Likewise to 001: $\frac{1}{3} \cdot \frac{4}{16} = \frac{1}{12}$; to 010: $\frac{1}{3} \cdot \frac{1}{13} = \frac{1}{39}$.
- Gibbs sometimes **redraws the value the variable already has**. Metropolis always moves to a heavier neighbour.
- Distance to p after 4 steps from 010: Metropolis 0.052, Gibbs 0.127. **Which sampler is faster depends on the model** (Peskun, 1973).

P. H. Peskun. Optimum Monte-Carlo sampling using Markov chains. *Biometrika* 60(3), 607–612, 1973.

A masked language model redraws one token given all the others



position 2 reads every other token



- A masked language model predicts $p_\theta(x_i | x_{-i})$ at any position; redrawing positions one by one is a **Gibbs-style chain**.
- **Our chain:** all three conditionals come from one table g , so they fit together: Gibbs samples exactly p .
- **A masked model:** each conditional is a separate network output. **There may be no joint distribution with all of them**, so the chain has no named target.
- **Fix:** score whole sentences with one energy $E(x)$; propose with the model's conditional; accept by Metropolis–Hastings. Target: $e^{-E(x)} / Z$.

Goyal, Dyer and Berg-Kirkpatrick, *ICLR* 2022.

Every chain we built is valid; the question now is how fast it gets there

1
Maximum
entropy

2
Moment
matching

3
Monte Carlo

4
Markov chains

5
Detailed
balance

6
Metropolis–
Hastings

7
Gibbs sampling

8
Mixing

WHAT WE HAVE

- Metropolis–Hastings and Gibbs: p is stationary.
- Every update is local and cheap, even with 2^{1024} states.

WHAT IS NOT GUARANTEED

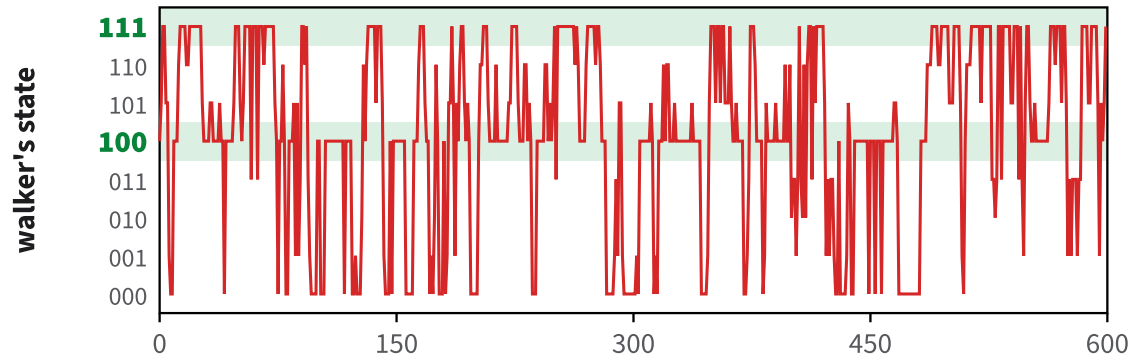
- The convergence theorem promises that the walkers reach p *eventually*, not how soon.
- Consecutive states are correlated, so N steps are not N independent draws.

NEXT QUESTION How fast does a chain forget its start, and what is a correlated sample worth?

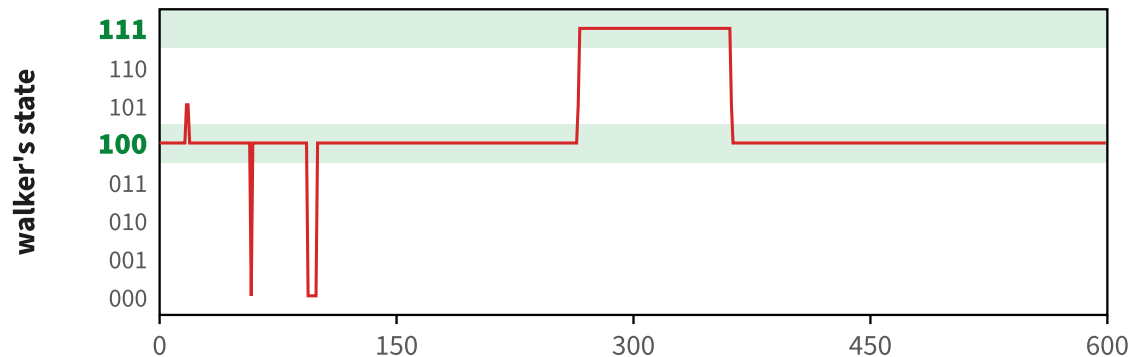
8 • Mixing

Cool the same chain: two peaks appear, and the walker gets stuck in one

T = 1: the walker visits every state



T = 0.2: long stays in the peaks 100 and 111



step t (red: the walker's state at each step; green rows: the two peaks)

- **How to read the left:** across, the step t ; up, the walker's state (one row per state). The red line shows where the walker is at each step.
- Target $p_T \propto e^{-E/T} = g^{1/T}$ (energy $E = -\ln g$); Metropolis moves as before.
- **Two heavy states**, 100 (weight 24) and 111 (20), **separated by light ones**: 101 (8) and 110 (5).
- $T = 1$: the walker **visits every state** within a few steps.
- $T = 0.2$: it **stays in one peak** for long stretches. **The chain is still valid; it is slow.**

Mixing time: how many steps until the start no longer matters

Definition (mixing time). With the distance used in the walker animations, $\|\mu - \pi\|_{\text{TV}} = \frac{1}{2} \sum_x |\mu(x) - \pi(x)|$,

$$t_{\text{mix}}(\varepsilon) = \min \left\{ t : \max_{x_0} \|P^t(x_0, \cdot) - \pi\|_{\text{TV}} \leq \varepsilon \right\},$$

where $P^t(x_0, \cdot)$ is the distribution after t steps from x_0 : **the worst start still ends within ε .**

Our chain, Metropolis, start 010. Distance 0.987, 0.731, 0.222, 0.118, 0.052, 0.028, 0.014, 0.008: **below 0.01 after 7 steps.**

Relaxation time. Late in a run, **each step multiplies the distance by the same factor $\lambda_2 < 1$:** the largest eigenvalue size of P after the eigenvalue 1, which belongs to π . Since $\lambda_2^t \approx e^{-t(1-\lambda_2)}$, the distance shrinks by $e \approx 2.7$ every $t_{\text{rel}} = 1/(1 - \lambda_2)$ steps. Fair coin: $\lambda_2 = \frac{2}{3}$, the factor we saw, $t_{\text{rel}} = 3$. Metropolis on our chain: $\lambda_2 = 0.8004$, $t_{\text{rel}} = 5$; indeed $0.8004^5 = 0.33$.

Both samplers converge geometrically, at the rate λ_2

Theorem 11 (convergence rate). Let P on a finite set be irreducible, aperiodic, and satisfy detailed balance with π ; let $\lambda_2 < 1$ be its largest eigenvalue size other than 1. From every start x_0 , for every t :

$$\|P^t(x_0, \cdot) - \pi\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\frac{1 - \pi(x_0)}{\pi(x_0)}} \lambda_2^t.$$

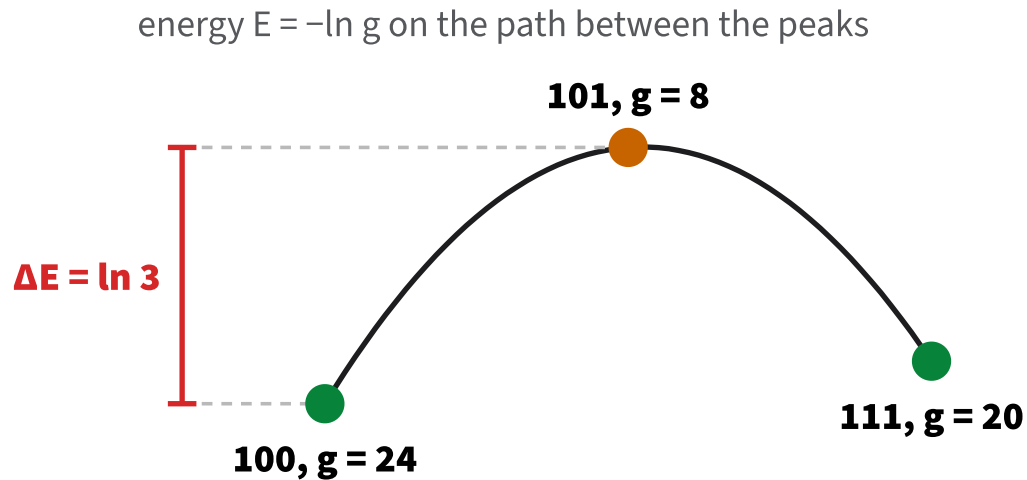
Metropolis–Hastings and random-scan Gibbs satisfy detailed balance, so it covers both samplers.

Proof idea. Detailed balance makes P symmetric after rescaling state x by $\sqrt{\pi(x)}$: real eigenvalues, perpendicular eigenvectors. Split the start into π (eigenvalue 1) plus other parts; **each step multiplies every other part by its eigenvalue**, so they shrink at least like λ_2^t . Cauchy–Schwarz converts this to total variation.

Our chain, start 010. $\frac{1}{2}\sqrt{77} = 4.39$. Metropolis: 4.39×0.8004^t ; Gibbs: 4.39×0.8514^t . The true distances lie far below: the bound gives the **rate**, not the constant.

P. Diaconis, D. Stroock. Geometric bounds for eigenvalues of Markov chains. *Ann. Appl. Probab.* 1(1), 36–61, 1991. Levin and Peres, *Markov Chains and Mixing Times*, 2nd ed., 2017, Ch. 12.

The waiting time grows exponentially in the height of the barrier over the temperature



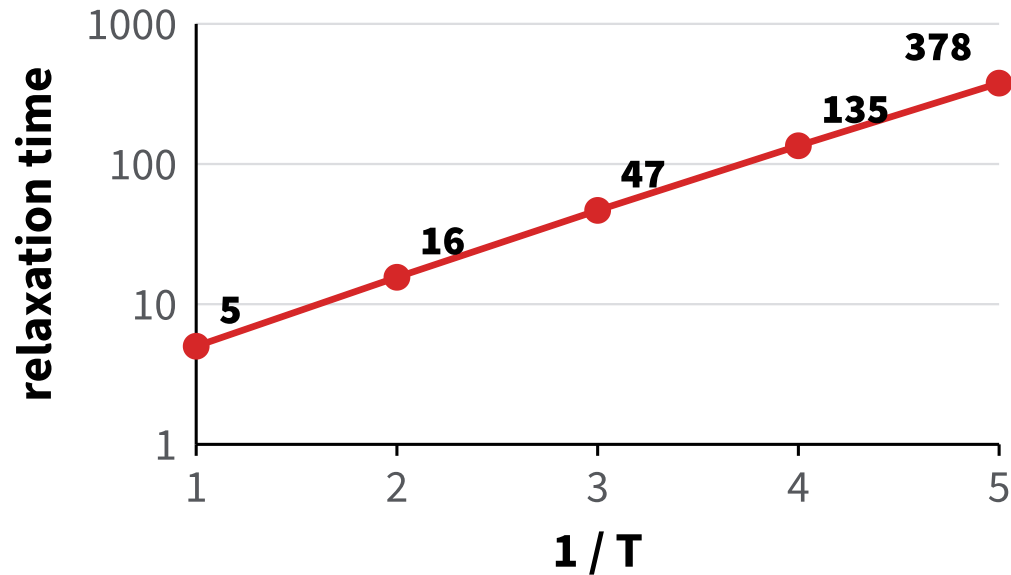
- From peak 100 (weight 24) to peak 111 (weight 20), the walker must pass a **lighter state**: 101 (weight 8) or 110 (weight 5). In energy $E = -\ln g$, a **hill** (top left).
- The target is $p_T \propto g^{1/T}$, so Metropolis accepts $100 \rightarrow 101$ with

$$\frac{p_T(101)}{p_T(100)} = \left(\frac{8}{24}\right)^{1/T} = e^{-\ln 3/T} :$$

$\frac{1}{3}$ at $T = 1$, but $\frac{1}{243}$ at $T = 0.2$.

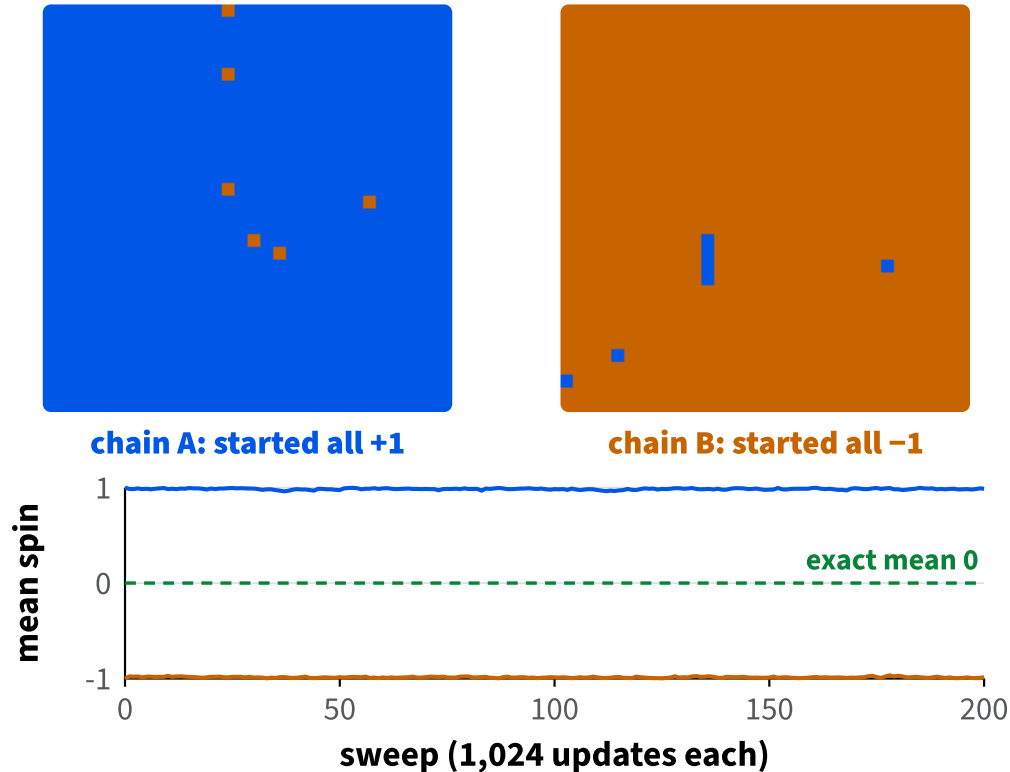
So the time to cross grows like $e^{\ln 3/T}$: **each time $1/T$ grows by 1, the waiting time multiplies by about 3.**

On our chain: each step down in temperature makes mixing about three times slower



- For each temperature, the **relaxation time** $t_{\text{rel}} = 1/(1 - \lambda_2)$, computed exactly from P : the steps the chain needs to forget its start.
- $1/T = 1, 2, 3, 4, 5$: **5, 16, 47, 135, 378 steps**. Each is about 3 times the one before ($16/5 = 3.2$, $378/135 = 2.8$).
- **How to read the plot:** across, $1/T$; up, the relaxation time on a **log scale** (each gridline 10 times the one below). A constant factor per step, here about 3, shows as a **straight line**.

Two chains, two different answers: the warning sign of poor mixing



$T = 1.5$ · sweep **200** · chain A average **0.986** · chain B average **-0.987**

- 32×32 grid, $h = 0$: by symmetry the exact mean spin (the average of the 1,024 spins) is 0.
- $T = 1.5$, below the critical temperature $T_c = 2.269$, where the grid has two modes (mostly +1 or mostly -1): chain A says +0.99, chain B -0.99. **Both confident, both wrong.**
- $T = 3$: both fluctuate around 0 and **agree.**

One chain at $T = 1.5$, 10,000 sweeps, a smooth trace. Trust its mean?

Yes

No

Answer. No: a stuck chain looks exactly like that. Only chains from different starts reveal it.

A walker repeats itself, so its steps carry less information than independent draws

60 independent draws from p : 29 switches



60 steps of one Metropolis walker: 10 switches

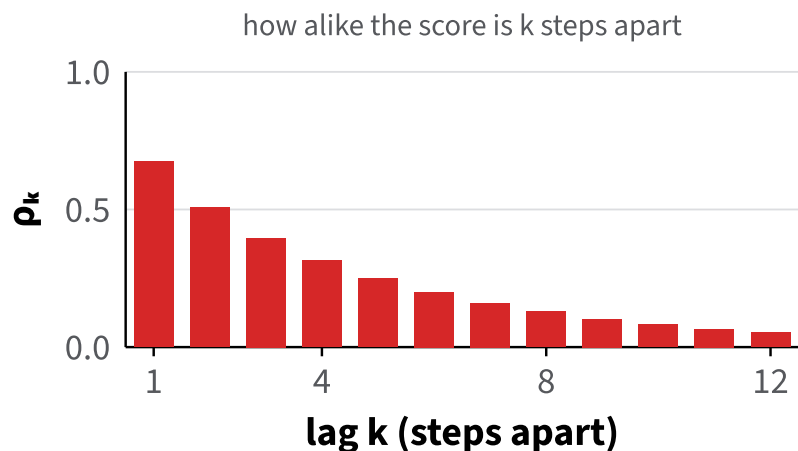


1 20 40 60
step t

■ score 1: $x_2 = 0$ ■ score 0: $x_2 = 1$

- We estimate $P(X_2 = 0) = 0.6154$ by averaging the **score** $g = 1$ if $x_2 = 0$, else 0.
- **Top:** independent draws, as in Monte Carlo: the score **switches often**.
- **Bottom:** one Metropolis walker. The score comes in **long runs**: a step changes one variable, or none.
- The walker is already at p ; the question is **how accurate its average is**. A run of equal scores is nearly one piece of evidence, repeated: on our chain, **the walker's 60 scores give about the accuracy of 8 independent draws** (the next slides compute this).

How accurate is an average along the walk? Its effective sample size



Definition. **Lag** k : the gap between steps t and $t + k$.

Autocorrelation $\rho_k = \text{Corr}(g(X_t), g(X_{t+k}))$: how alike the score is k steps apart (0: no memory; 1: identical). Left: our walker at p .

Independent draws: $\text{Var}(\hat{\mu}_N) = \sigma^2/N$, since variances add. Along the walk, **the covariances add too**:

$$\text{Var}(\hat{\mu}_N) = \frac{\sigma^2}{N^2} \sum_{s,t=1}^N \rho_{|s-t|} \approx \frac{\sigma^2}{N} \tau, \quad \tau = 1 + 2 \sum_{k \geq 1} \rho_k.$$

- τ : **how many walker steps are worth one independent draw.** So N steps count as $N_{\text{eff}} = N/\tau$, the **effective sample size**, and the error is $\sigma/\sqrt{N_{\text{eff}}}$.
- Independent draws: every $\rho_k = 0$, $\tau = 1$. A stuck walker: τ **huge**, N_{eff} **tiny**.

On our chain, 10,000 Metropolis steps are worth about 1,372 independent draws

$g = 1[x_2 = 0]$ along the Metropolis chain at $T = 1$, computed exactly from P :

$$\rho_1 = 0.675, \quad \rho_2 = 0.508, \quad \rho_3 = 0.396, \quad \dots \quad \tau = 1 + 2 \sum_{k \geq 1} \rho_k = 7.29.$$

$$N_{\text{eff}} = \frac{10,000}{7.29} = 1,372, \quad \text{error} = \frac{0.4865}{\sqrt{1372}} = 0.0131 \quad \text{instead of} \quad \frac{0.4865}{\sqrt{10,000}} = 0.0049.$$

- Gibbs on the same chain: $\tau = 10.33$, **only 968 effective draws.**
- **Report $\sigma / \sqrt{N_{\text{eff}}}$, never $\sigma / \sqrt{N}$, for a Markov chain.**

Four checks before trusting a Markov chain average

Check	What to do	Warning sign	Seen today
1. Burn-in	leave the first steps out of the average	early steps still sit near the start	walkers started at 010 came within 0.01 of p only after 7 steps
2. Trace plot	plot the score $g(X_t)$ against t	long flat stretches	the walker at $T = 0.2$ stays in one peak
3. Several chains	start them far apart and compare	the chains disagree	grid at $T = 1.5$: +0.99 against −0.99
4. Effective sample size	estimate τ ; report $\sigma / \sqrt{N_{\text{eff}}}$	N_{eff} tiny	10,000 steps worth 1,372 draws

Why burn-in: the start is our choice, not a draw from p (010 has $p = \frac{1}{78}$), so the first steps over-represent it.

$\hat{R}$ puts check 3 into one number: how far apart the chains' averages are, compared with how much each chain varies on its own. Chains that agree give $\hat{R} \approx 1$. The grid chains at $T = 1.5$ average +0.99 and −0.99 while each barely moves: $\hat{R}$ far above 1.

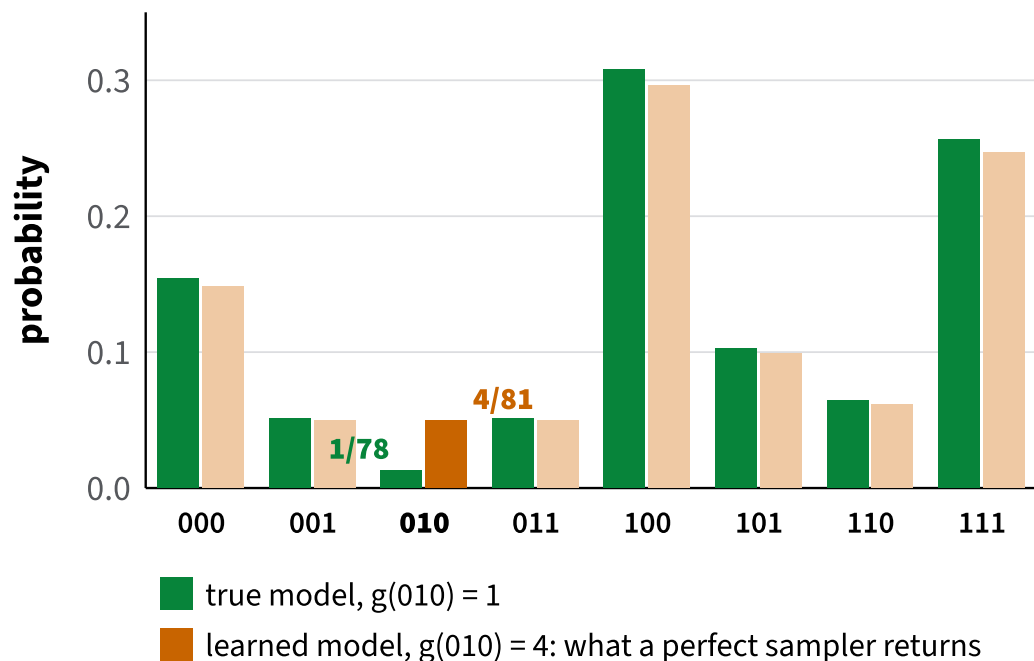
A. Gelman, D. B. Rubin. Inference from iterative simulation using multiple sequences. *Statistical Science* 7(4), 457–472, 1992.

Three kinds of chains, and how to tell them apart

kind	example from today	symptom	does running longer help?
invalid	uneven proposal without Hastings' ratio	converges to the wrong π : $P(X_2=0) = 0.150$	no
valid but slow	chain at $T = 0.2$; grid at $T = 1.5$	chains disagree; τ huge	yes, eventually
well mixed	chain at $T = 1$	relaxation 5 steps, $\tau = 7.3$	error $\sigma/\sqrt{N_{\text{eff}}}$

- **Invalid chains** are caught by checking detailed balance on a few edges, or by a model small enough to enumerate, like ours.
- **Slow chains** are caught by several chains from spread-out starts.

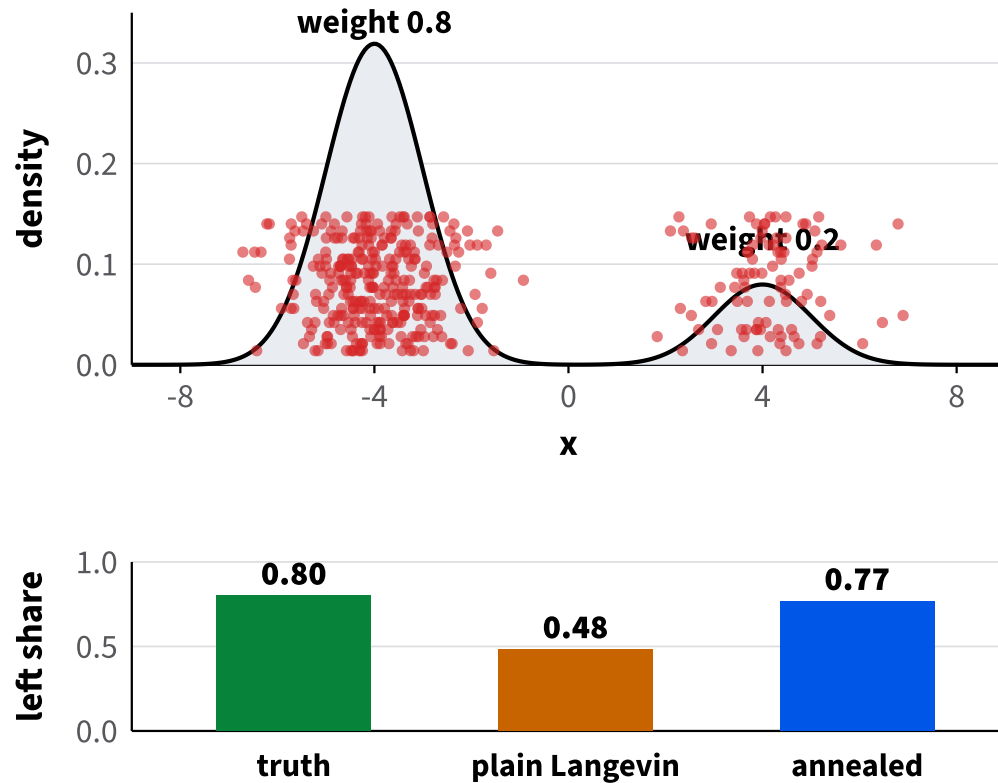
A perfect sampler reproduces the model, including the model's mistakes



- A sampler reproduces **whatever model it is given**. Suppose a learned model had $g(010) = 4$ instead of 1: its energy $-\ln 4$ is **too low**.
- A perfect Metropolis walker then visits 010 with $\frac{4}{81} = 0.049$ instead of $\frac{1}{81} = 0.013$: **almost 4 times too often**. The chain is valid; the model is wrong.
- **A real case:** learned energies for atoms came out too low on rare configurations, such as a missing atom (Deng et al., 2025); samplers then visit them too often.
- **A valid sampler and a valid model are separate questions:** check the model against reference data.

B. Deng, Y. Choi, P. Zhong et al. Systematic softening in universal machine learning interatomic potentials. *npj Computational Materials* 11, 9, 2025.

Slow mixing between modes is why diffusion models add noise at many scales



- **Target** (top): two bumps; the left one holds **80% of the probability**.
- **Langevin dynamics**: $x \leftarrow x + \varepsilon \nabla \ln p(x) + \sqrt{2\varepsilon} \xi$, ξ standard normal: drift toward higher probability, plus noise. No Z needed.
- **Plain**: walkers start half on each side; each stays in its nearest bump, **almost never crossing the gap**. About half end on the left (bottom), not 0.8: our chain's trap at $T = 0.2$.
- **Annealed**: add heavy noise so the bumps **merge**, then lower it step by step: about 0.8 end on the left. Score-based diffusion models sample this way.

Y. Song, S. Ermon. Generative modeling by estimating gradients of the data distribution. *NeurIPS* 2019.

Correlated evaluation questions shrink the effective number of questions

- **Background:** a benchmark score is an average of N 0/1 scores, and its error bar $\sqrt{a(1-a)/N}$ assumes **independent questions**.
- Many benchmarks ask several questions about one passage. A model that misreads the passage tends to miss them all: **answers in a group move together**, like consecutive walker steps.
- **Main idea:** groups of m questions, with correlation ρ between two answers of a group, are worth $N_{\text{eff}} = N / (1 + (m - 1)\rho)$ independent questions: Kish's **design effect**.

$$N = 1000, m = 10, \rho = 0.3 : \quad N_{\text{eff}} = \frac{1000}{1 + 9 \times 0.3} = \frac{1000}{3.7} = 270.$$

At accuracy 0.80 the standard error grows from 0.0126 to $0.0126 \times \sqrt{3.7} = 0.0243$: the 95% interval widens from ± 2.5 to ± 4.8 points. **Same fix as for chains:** compute the error from the effective sample size (clustered standard errors).

L. Kish. *Survey Sampling*. Wiley, 1965. E. Miller. Adding error bars to evals. arXiv:2411.00640, 2024.

Sampled model averages let us fit models without Z

1

Maximum entropy

2

Moment matching

3

Monte Carlo

4

Markov chains

5

Detailed balance

6

Metropolis–Hastings

7

Gibbs sampling

8

Mixing

WHAT WE HAVE

- Any model average $\mathbb{E}_\theta[\phi_k]$, from a valid chain, with an error bar $\sigma/\sqrt{N_{\text{eff}}}$.
- So the fitting rule $\theta_k \leftarrow \theta_k + \eta (m_k - \mathbb{E}_\theta[\phi_k])$ runs on grids of any size.

WHAT COMES NEXT

- Some variables are never observed, such as a machine's hidden state or a topic.
- Then even the *data* average needs inference.

LECTURE 5 How do we fit a model when some variables are hidden?

Expectation–maximization

The story in one picture: one formula per question

1 • Maximum entropy

Known averages, which distribution?

$$p_{\theta} \propto e^{\sum_k \theta_k \phi_k}$$



2 • Moment matching

Which θ ?

$$\partial_{\theta_k} \ell = m_k - \mathbb{E}_{\theta}[\phi_k]$$



3 • Monte Carlo

Average without the sum?

error $\sigma / \sqrt{N}$



4 • Markov chains

Draw without Z ?

$$\mu_{t+1} = \mu_t P \rightarrow \pi$$

5 • Detailed balance

Does it end at p ?

$$\pi(x)P(x, x') = \pi(x')P(x', x)$$



6 • Metropolis–Hastings

A recipe?

$$\min\left\{1, \frac{g(x')q(x|x')}{g(x)q(x'|x)}\right\}$$



7 • Gibbs sampling

On a factor graph?

$$x_i \sim p(x_i \mid \text{blanket})$$



8 • Mixing

Can we trust it?

$$\sigma / \sqrt{N_{\text{eff}}}, N_{\text{eff}} = N / \tau$$

Two ideas avoid Z : **replace sums by averages** (Monte Carlo, with error bars) and **use only ratios** (the Markov chain samplers). Both return in every later week.

Three research questions you can start this week with today's tools

1. Power sampling on a toy language model

Question. How many Metropolis–Hastings steps does p^α need as α grows?

First experiment. Read our chain as a 3-token model. Compare token-level temperature with Metropolis–Hastings on p^α for $\alpha = 1, 2, 4, 8$; plot the distance to the exact target against steps.

Start from Karan and Du (2025).

2. Where is Gibbs sampling slowest?

Question. How does the autocorrelation time τ of the mean spin depend on T and on the grid size L ?

First experiment. Measure τ for $L = 8, 16, 32$ and T from 1.5 to 3.5. Locate the peak and compare it with $T_c = 2.269$.

Cluster moves fix it: U. Wolff, Collective Monte Carlo updating for spin systems, *Phys. Rev. Lett.* 62(4), 361–364, 1989.

3. Is my sampler valid?

Question. When conditionals are learned separately, is there a joint distribution behind them?

First experiment. Perturb one entry inside the conditional of X_2 only. Test detailed balance edge by edge; find the stationary distribution; repair it with a Metropolis–Hastings step.

Start from Goyal, Dyer and Berg-Kirkpatrick (2022).

Three questions to answer before Lecture 5

1. Symmetric proposal, $g(x') = 3 g(x)$. Acceptance of the move, and of the reverse move?

1 and 1/3

1/3 and 1

3/4 and 1/4

Answer. 1 and $\min(1, \frac{1}{3}) = \frac{1}{3}$: ratio 3 on the edge, detailed balance.

2. A Markov chain runs four times longer. Its error?

Divided by 4

Halved, if τ is unchanged

Unchanged

Answer. Halved: error $\sigma / \sqrt{N/\tau}$. A stuck chain can have τ longer than any run.

3. 70% of proposals accepted, but the chain never leaves its peak. Reliable?

Yes, 70% is high

No

Answer. No. Acceptance measures local moves, not travel between peaks.

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The chain, the eight-state cube, the walkers and all worked numbers are original teaching constructions, computed exactly by the course program.